# LDSC Estimator Variant Comparison

All six free/fixed-intercept × weighted/unweighted × two-step/single-step LDSC variants, compared together, under real chr1/chr22 LD structure

## Executive Summary

Every simulation in this project's realistic-LD pipeline saves six LDSC heritability estimator variants per replicate, varying three binary choices: intercept (free vs. fixed at 1), SNP weighting (the standard LDSC weights vs. uniform weights), and the two-step procedure (which first estimates the intercept excluding high-chi-square SNPs, then re-estimates the slope - vs. a single regression step using all SNPs and both parameters free). This report compares all six together, across every scale (n=500/1000/2000/5000), chromosome (chr1, chr22), and the project's four core scenarios (reference panel present or not, stratification vector f shared with the reference panel or drawn independently).

**Headline finding:** in three of the four scenarios (no reference panel with either f-sharing choice, and a reference panel with *shared* f), all six variants stay within roughly a factor of 2 of each other and of h2_pop across the full $F_{ST}$ range. But in the pathological scenario identified in the companion report - a reference panel whose stratification vector f is *mismatched* with the main sample - the six variants **split into two completely different failure modes**: the two-step variants (the LDSC default) inflate dramatically, reaching 4-6× h2_pop by $F_{ST}$=0.5 at the largest scale, while the two single-step variants instead **collapse toward zero** (down to ~0.005, roughly 1.6% of h2_pop). The two unweighted two-step variants sit in between, rising only modestly to ~0.3-0.7.

This means the two-step procedure - not the free/fixed intercept choice, and not the weighting scheme - is the single largest driver of whether an f-mismatched reference panel causes estimates to explode or collapse. Disabling it does not fix the estimator; it flips the direction of the failure.

# Methods

See the companion report (realistic_LD_sharedf_vs_indepf_summary.pdf) for full simulation methodology. The six LDSC variants compared here, all computed from the same per-replicate data (same X, y, LD scores):

- **h2_ldsc_reg**: Free intercept, weighted, two-step
- **h2_ldsc_fixed**: Fixed intercept, weighted, two-step
- **h2_ldsc_fixed_nw**: Fixed intercept, unweighted, two-step
- **h2_ldsc_nw**: Free intercept, unweighted, two-step
- **h2_ldsc_nts**: Free intercept, weighted, single-step
- **h2_ldsc_nwts**: Free intercept, unweighted, single-step

All panels use a linear y-axis. The grey dashed line is h2_pop=0.3, the true simulated genetic heritability. The black dotted line is the theoretical total h2 (h2_pop + sigma_s$^2$/C_f, C_f = mean(f$^2$/(1+f$^2$))), using that scenario's own per-replicate f draws - the same formula validated against the project's R implementation in the companion report. Note that in the pathological reference-panel/independent-f scenario, the value range spans roughly 0.001 to above 6; on a linear axis the single-step variants (which collapse toward zero) can appear visually flat against the baseline next to the two-step variants (which inflate well above h2_pop).

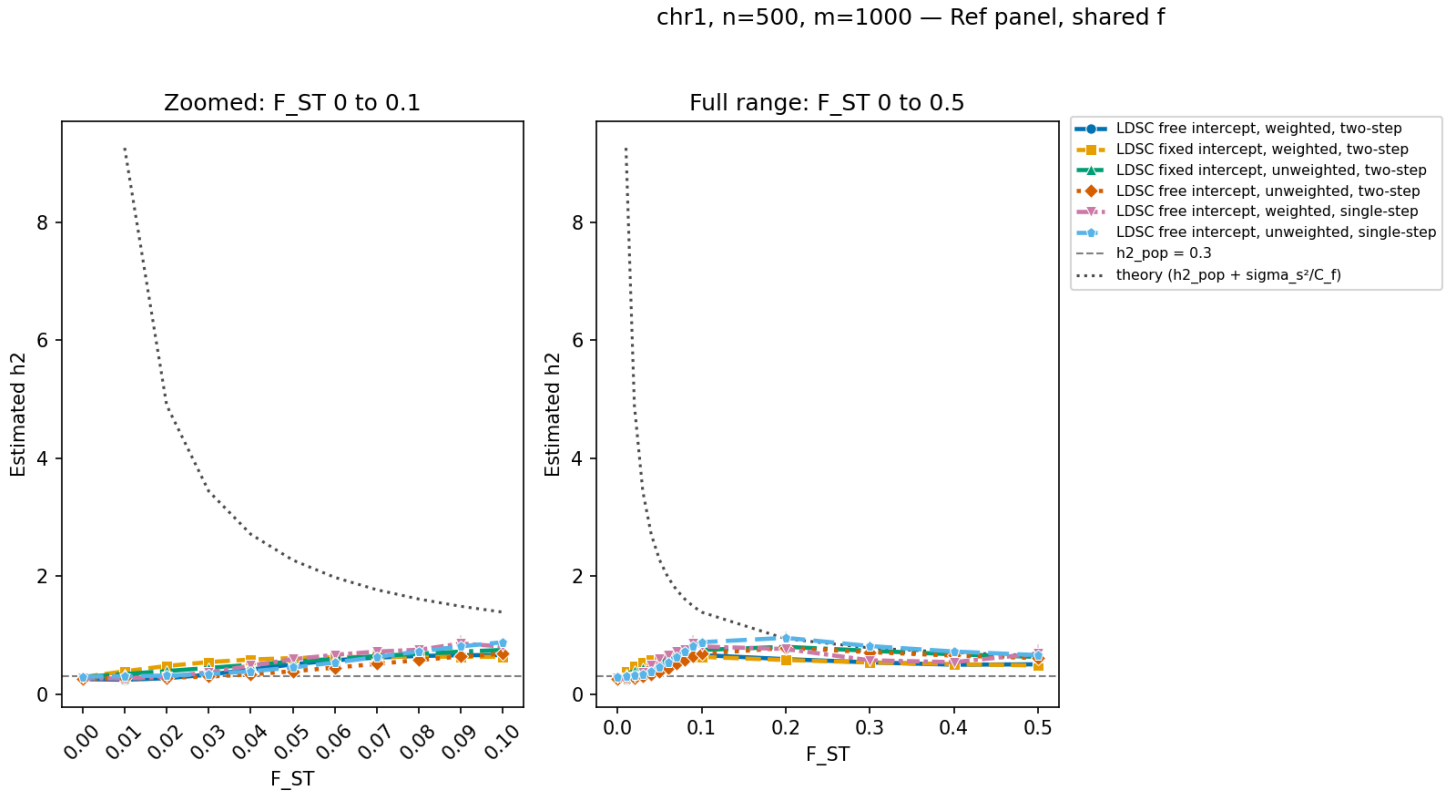
# Summary: All Six Estimators at F_ST=0.5, Reference Panel + Independent f

Mean h2 estimate (100 replicates) for each of the six LDSC variants at $F_{ST}$=0.5, in the scenario where the divergence is largest: a reference panel whose f does not match the main sample's.

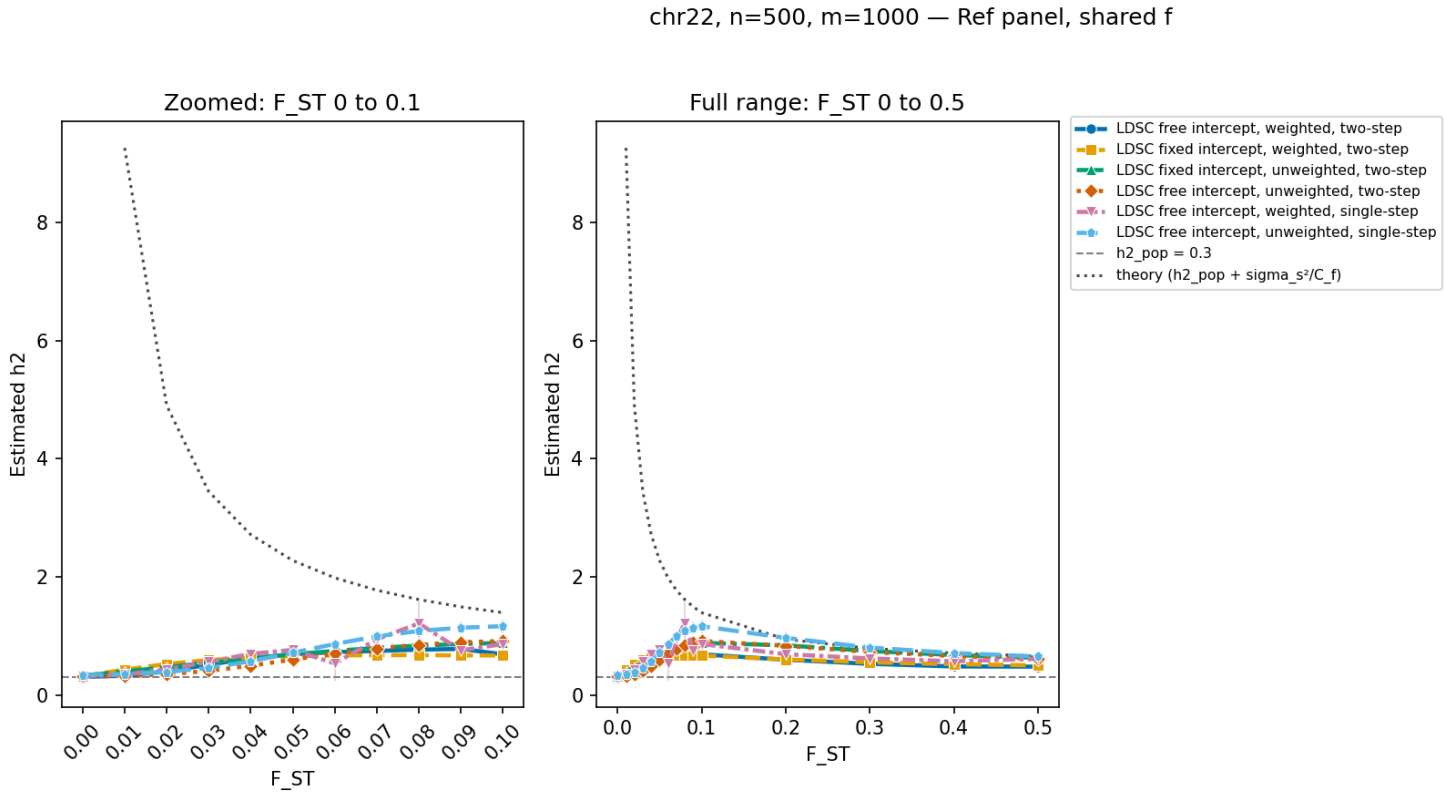
| Chrom | n | Free/Wt/2step | Fixed/Wt/2step | Fixed/Unwt/2step | Free/Unwt/2step | Free/Wt/1step | Free/Unwt/1step |
|---|---|---|---|---|---|---|---|
| chr1 | 500 | 0.869 | 2.560 | 0.347 | 0.225 | 0.0233 | 0.0150 |
| chr1 | 1000 | 1.512 | 3.205 | 0.320 | 0.250 | 0.0083 | 0.0031 |
| chr1 | 2000 | 2.532 | 4.301 | 0.304 | 0.265 | 0.0052 | 0.0002 |
| chr1 | 5000 | 4.529 | 6.037 | 0.348 | 0.332 | 0.0047 | 0.0053 |
| chr22 | 500 | 0.967 | 3.005 | 0.333 | 0.216 | 0.0254 | 0.0135 |
| chr22 | 1000 | 1.478 | 3.314 | 0.319 | 0.247 | 0.0151 | 0.0055 |
| chr22 | 2000 | 2.481 | 4.315 | 0.309 | 0.270 | 0.0046 | 0.0003 |
| chr22 | 5000 | 4.394 | 5.865 | 0.348 | 0.331 | 0.0054 | 0.0062 |

The two two-step, weighted/fixed-intercept-adjacent variants (columns 1-2) inflate sharply with scale (0.5-0.9 at n=500 up to 4.4-6.0 at n=5000); the two single-step variants (columns 5-6) collapse toward zero instead (note the 4-decimal precision needed - these are two orders of magnitude smaller); the unweighted two-step variants (columns 3-4) stay modest throughout.

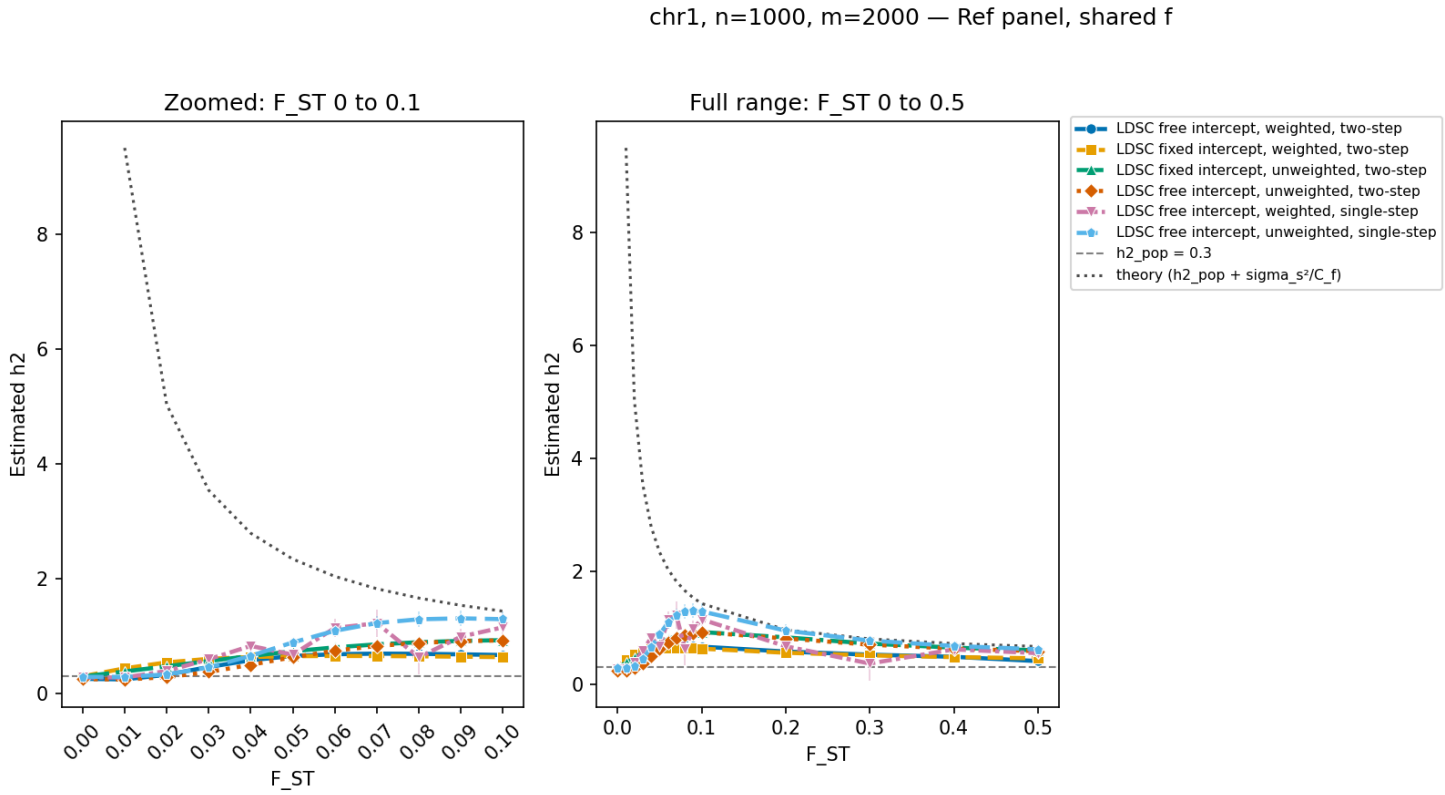
# Figures: Ref panel, shared f



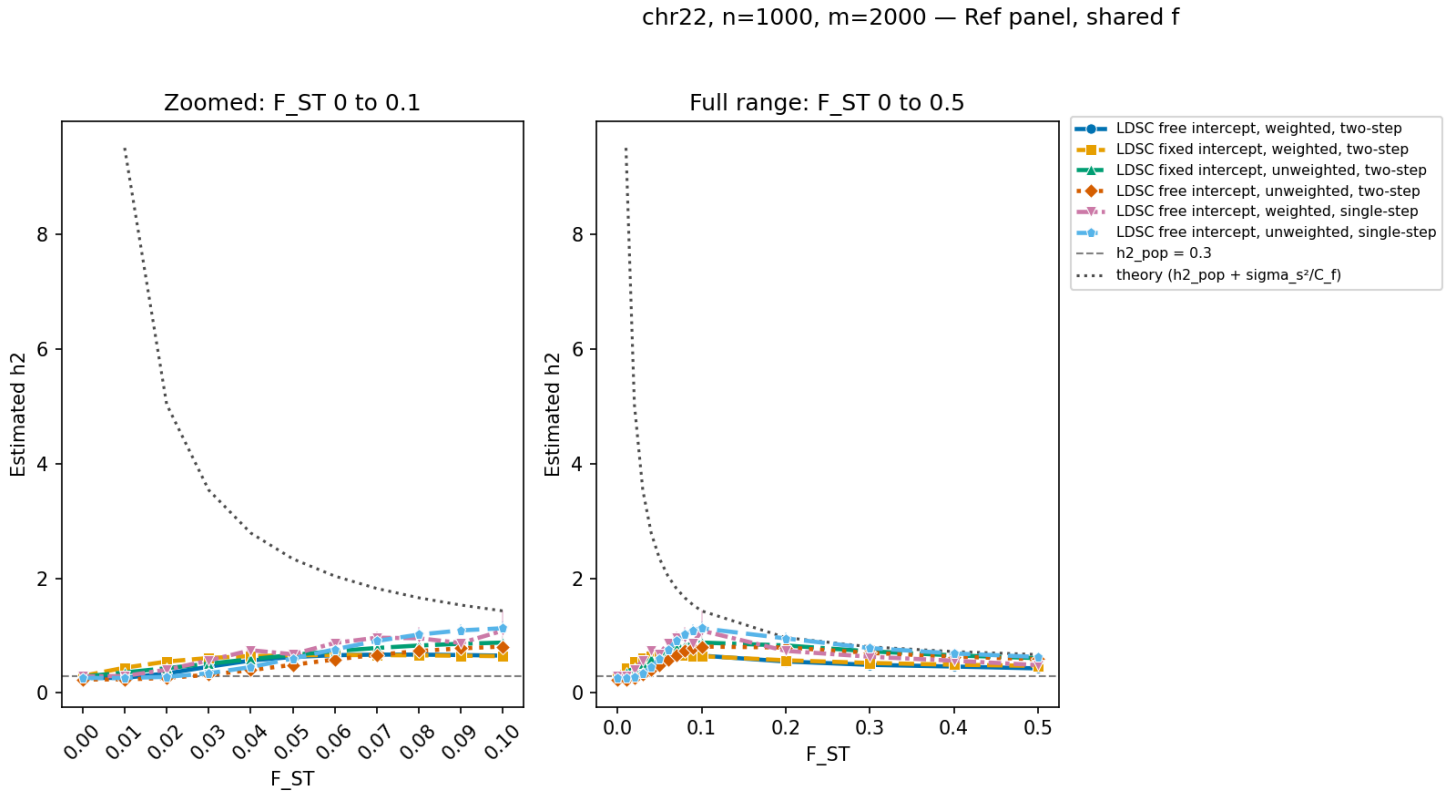
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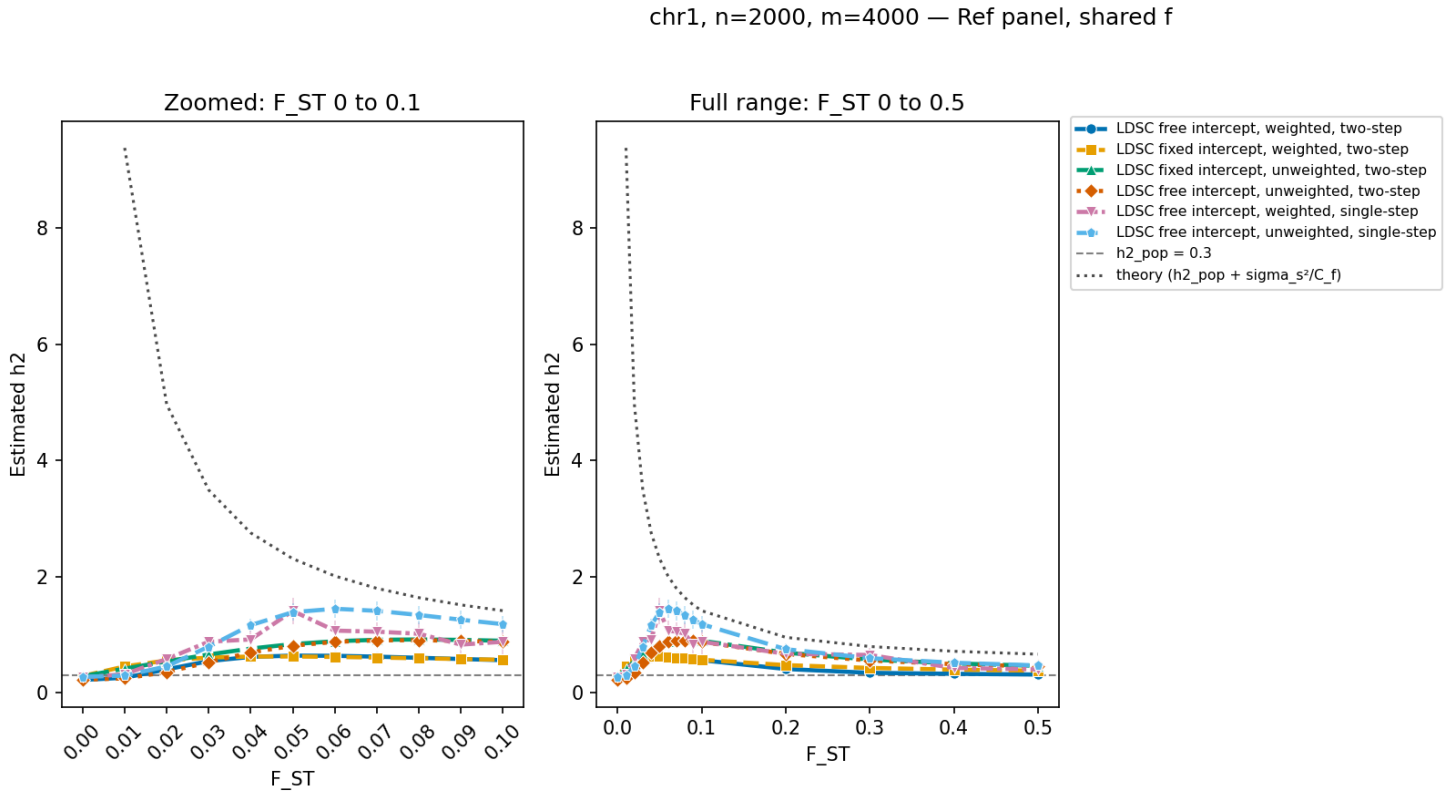
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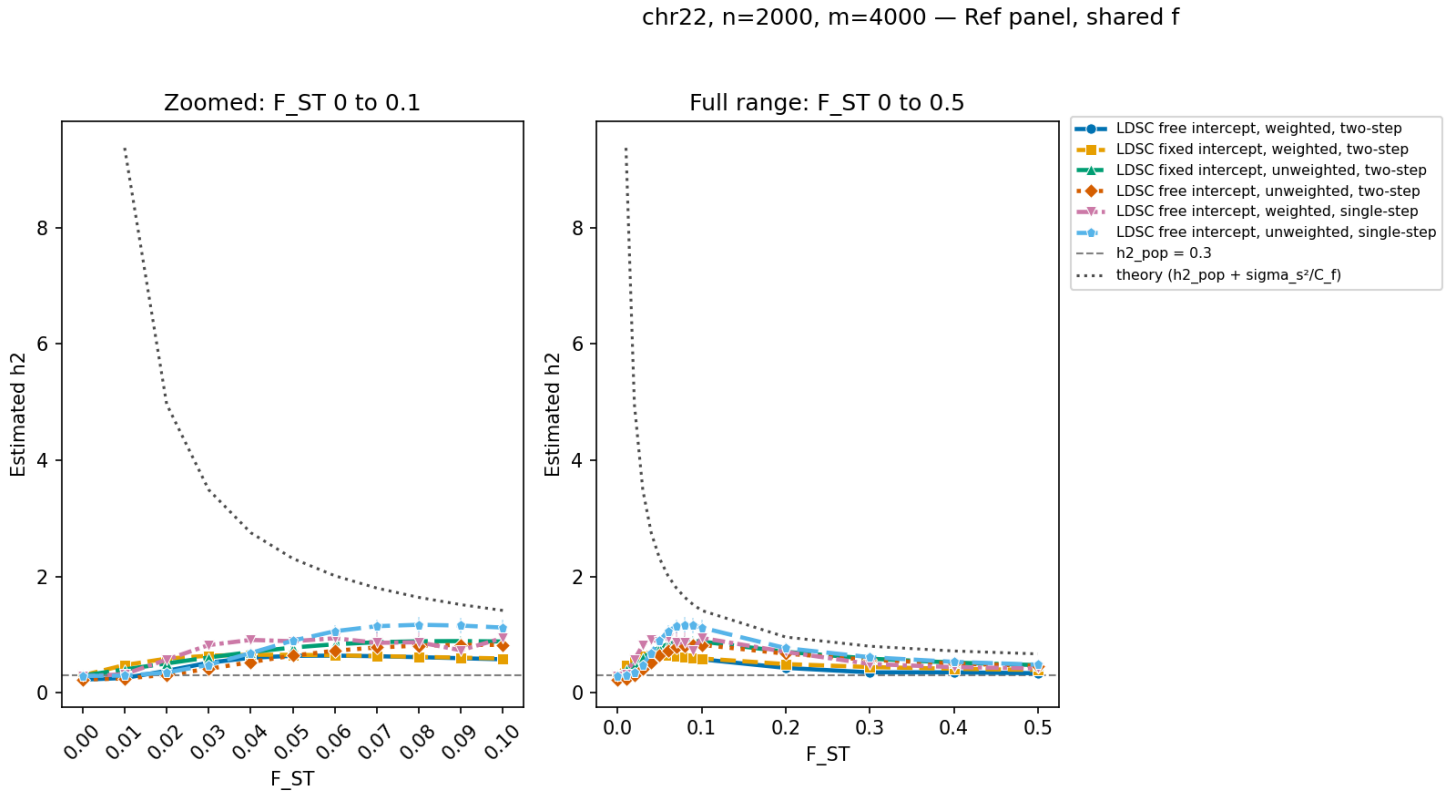
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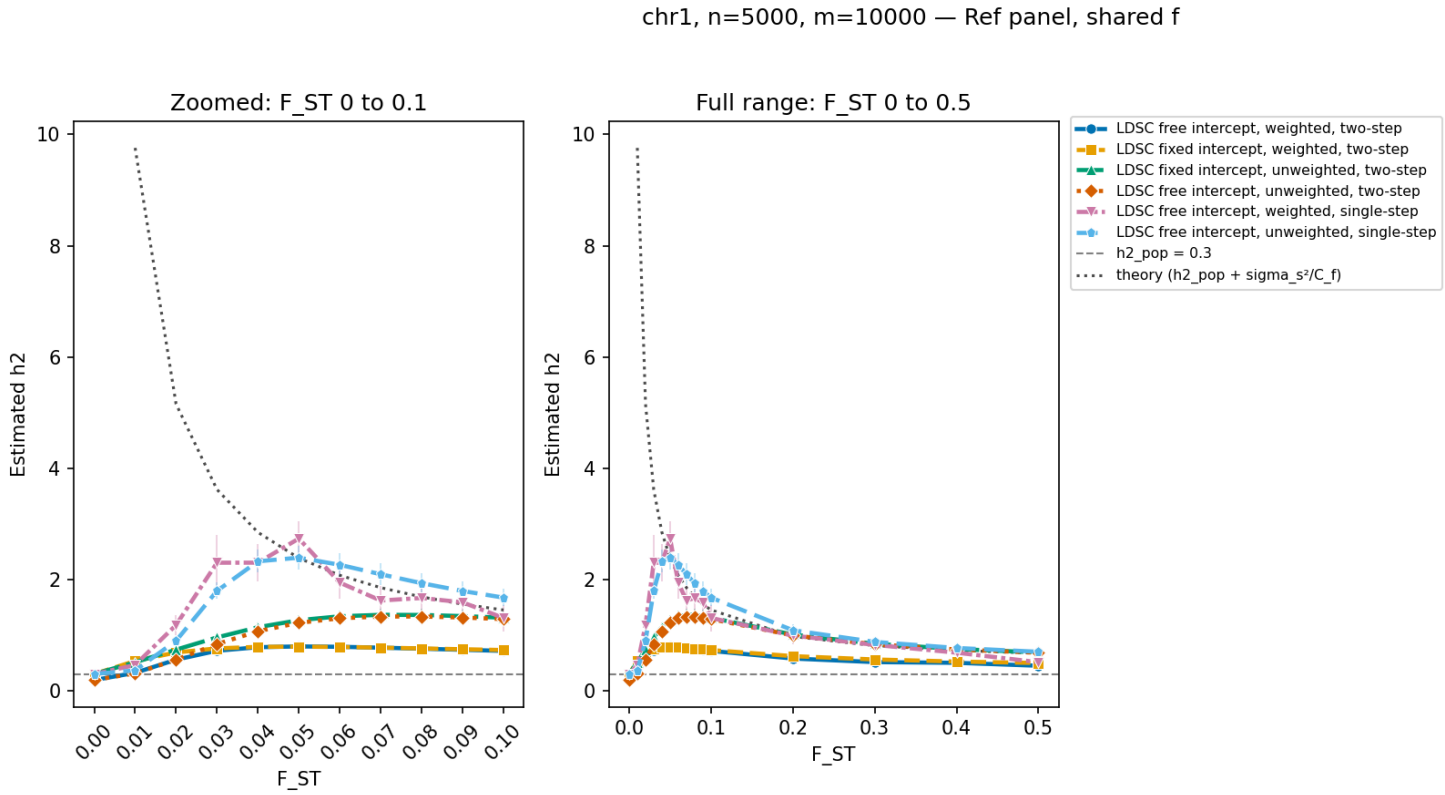
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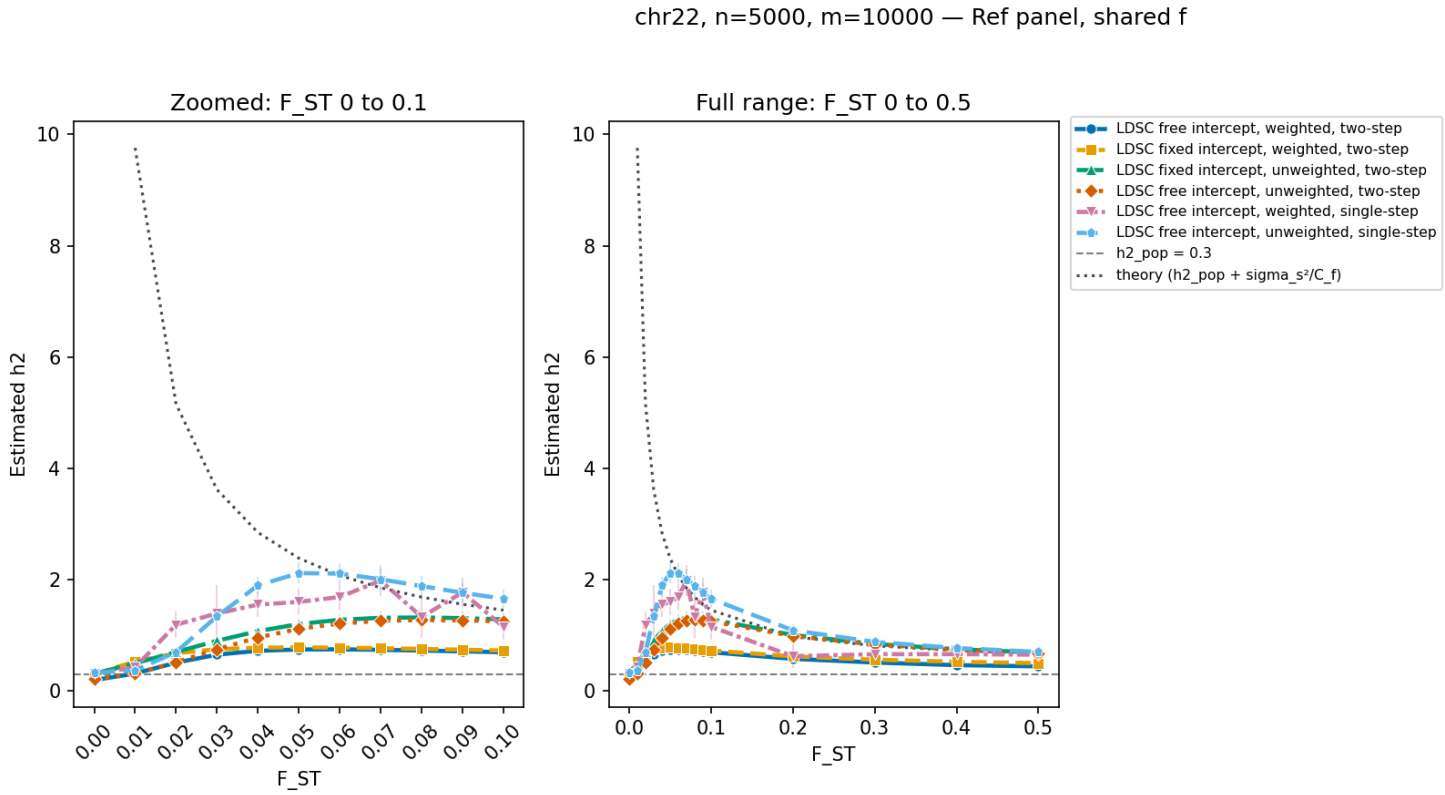
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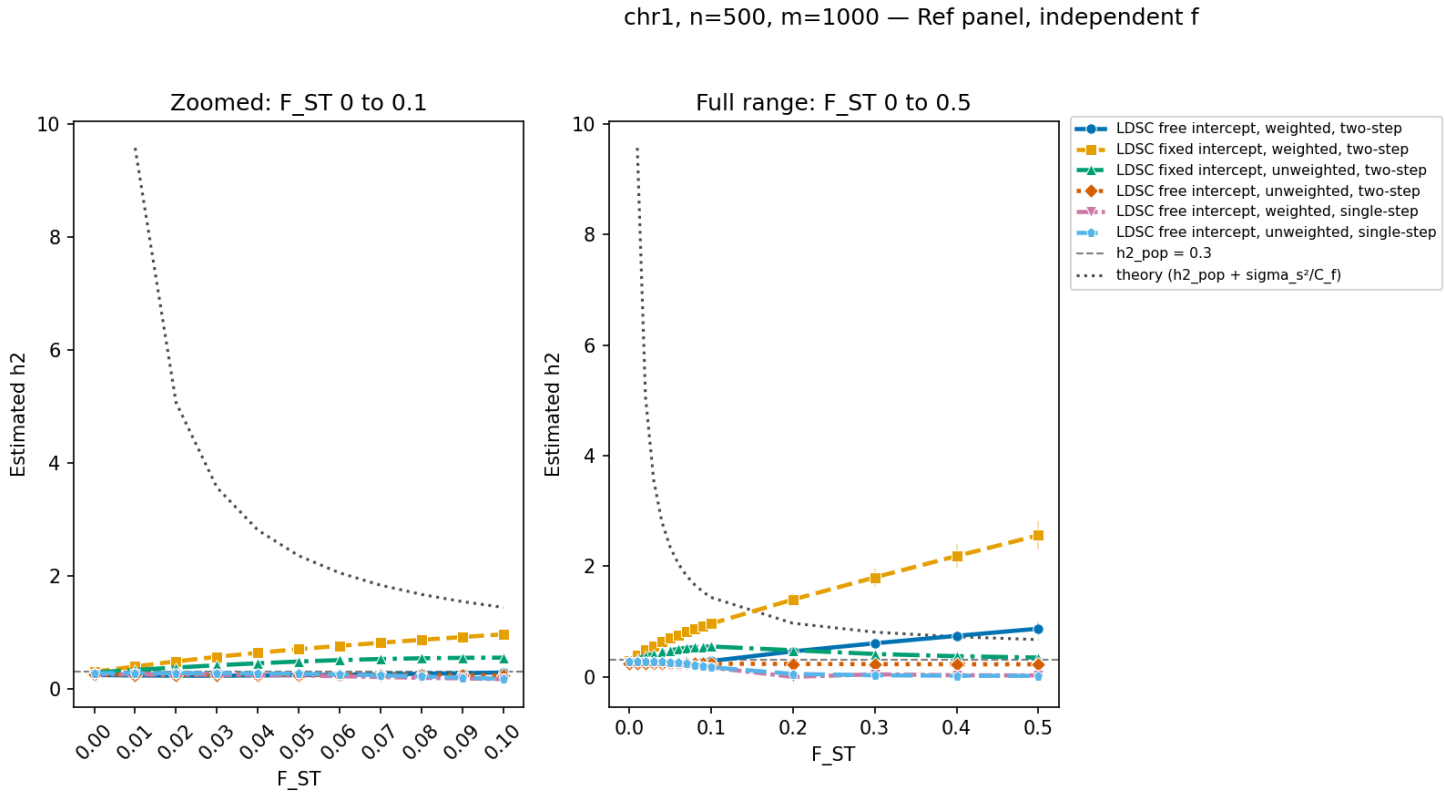
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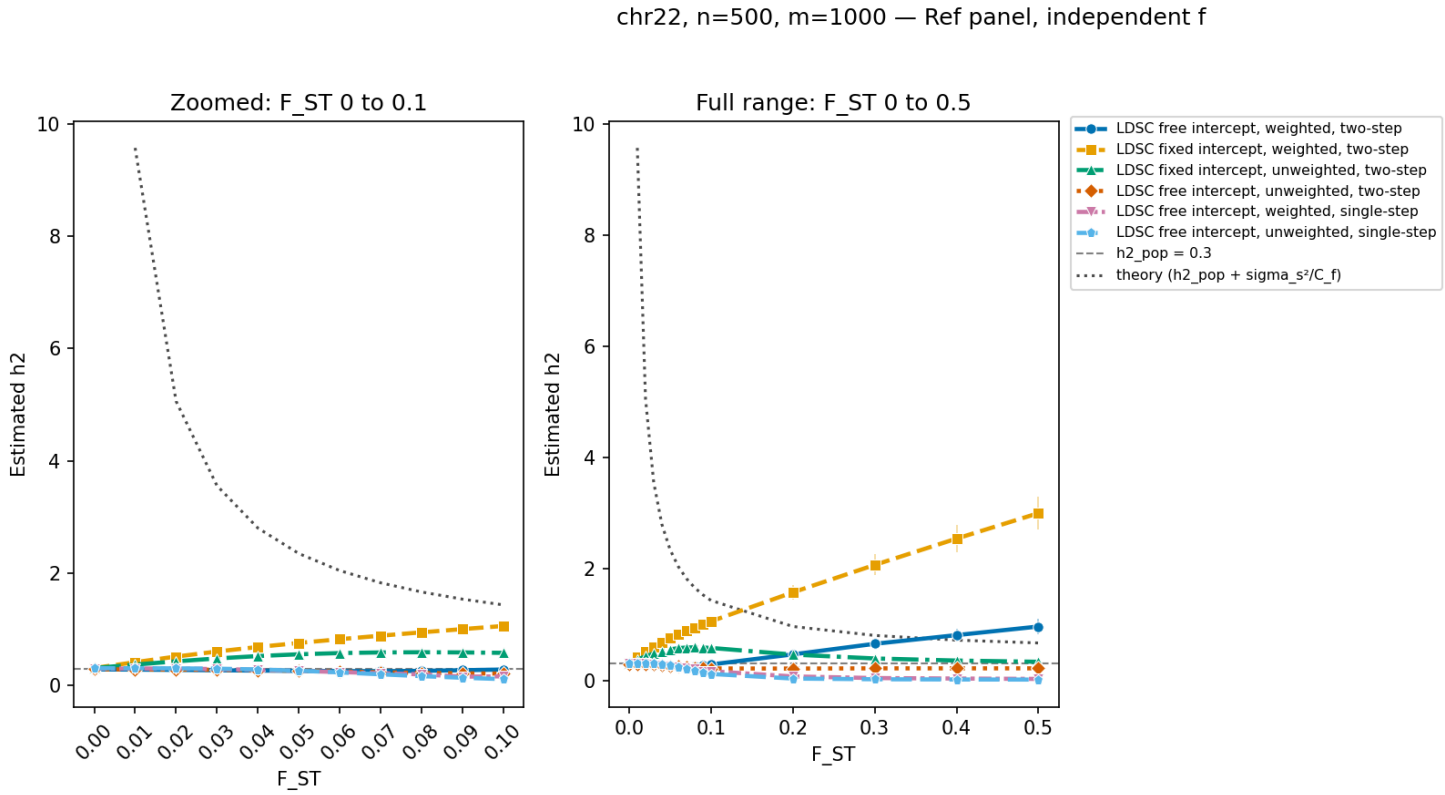
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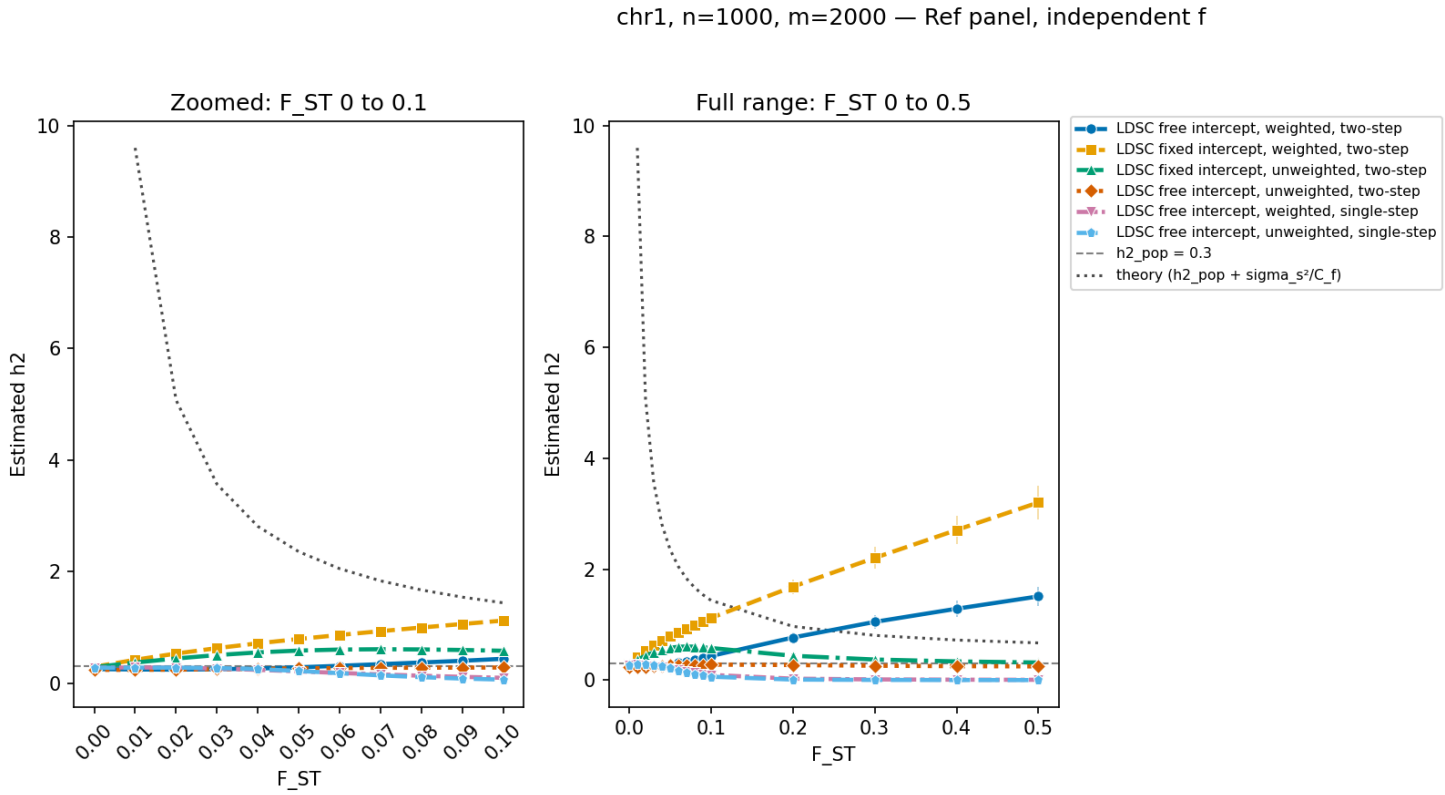
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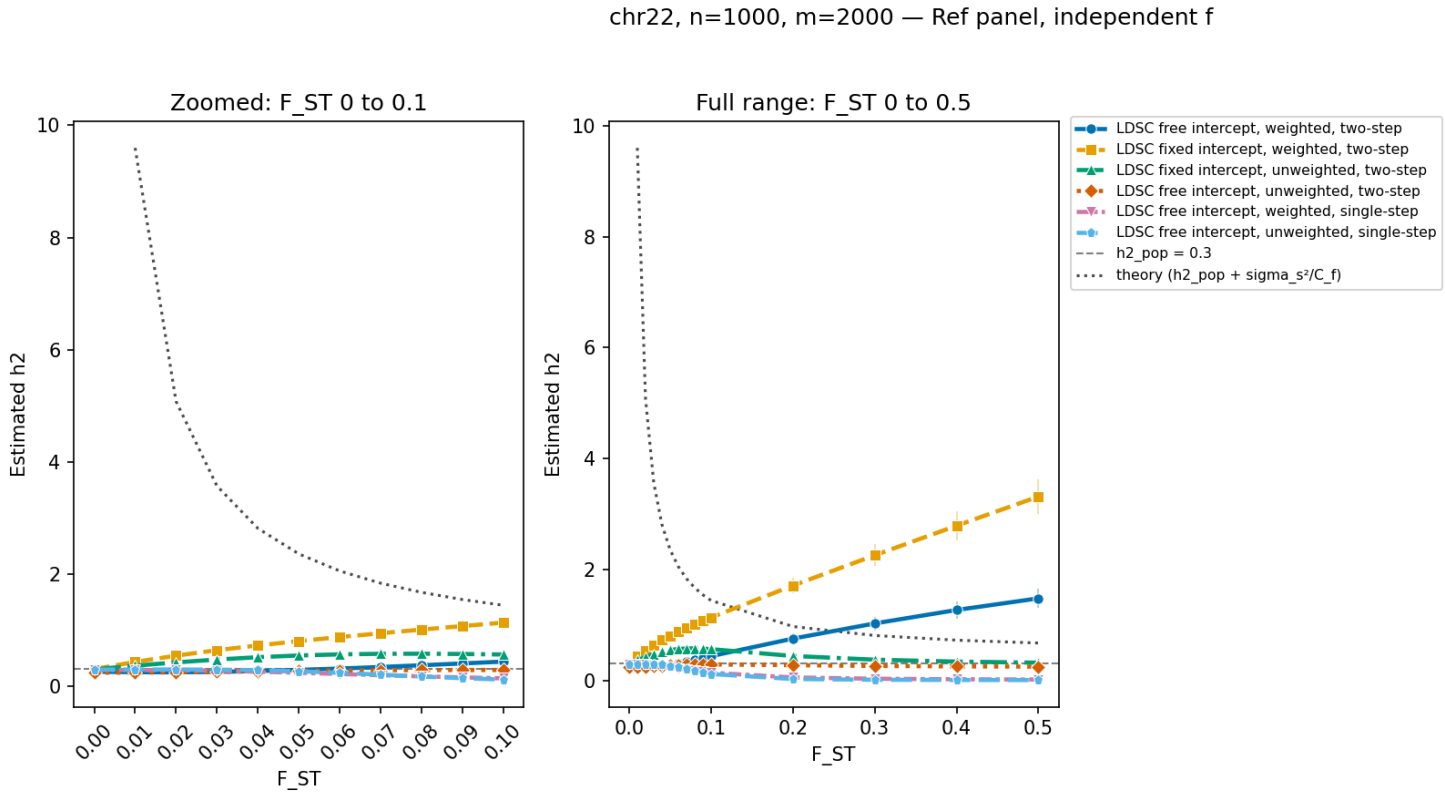
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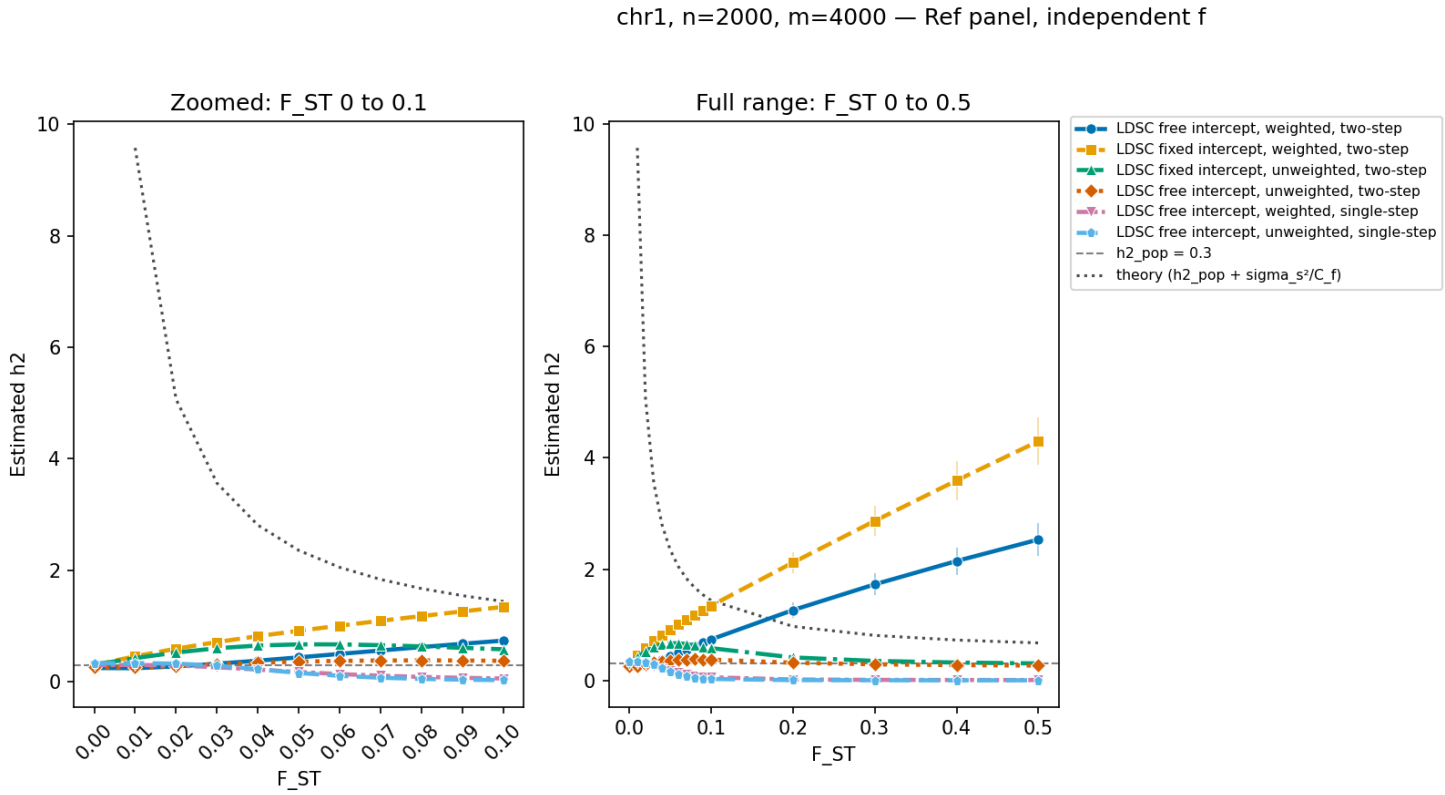
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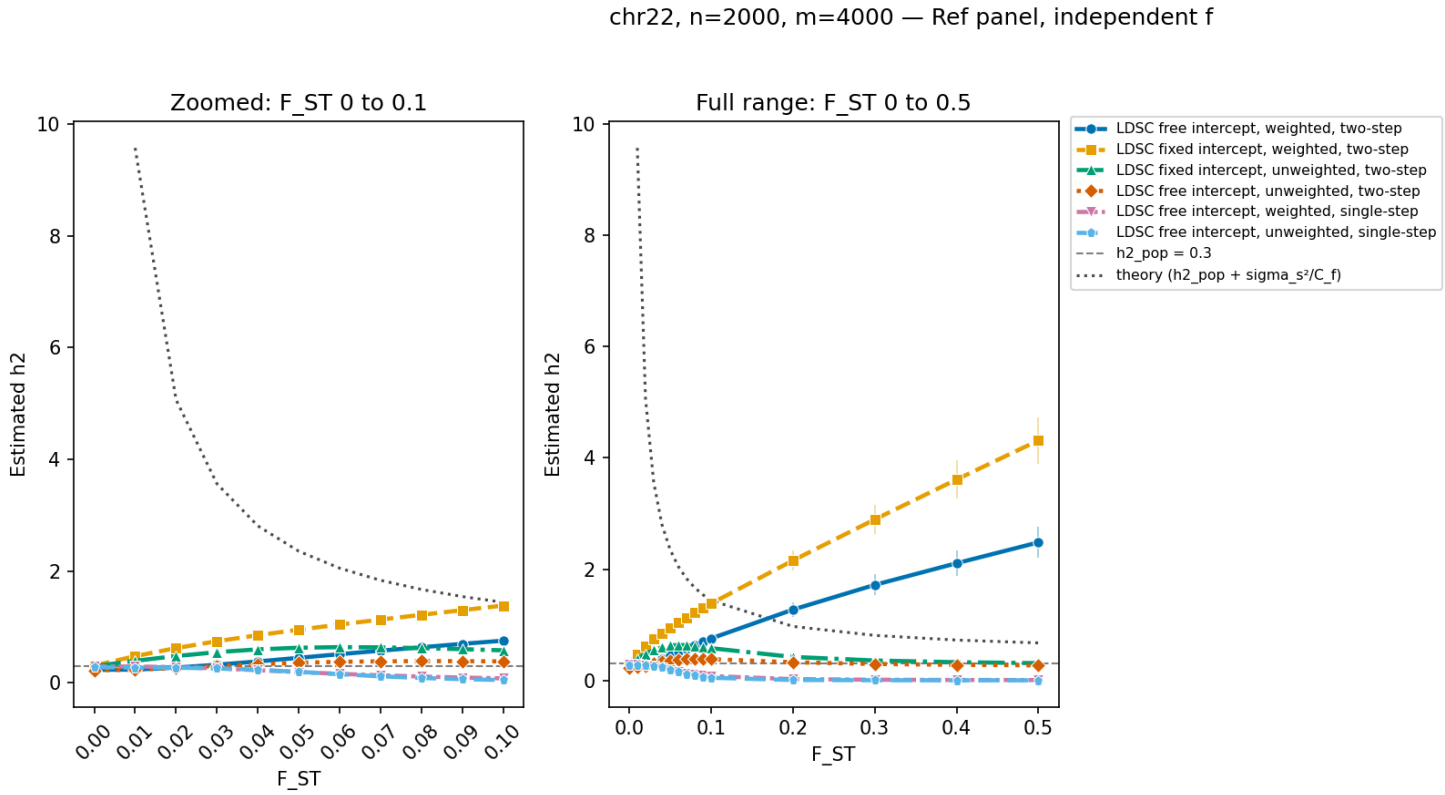
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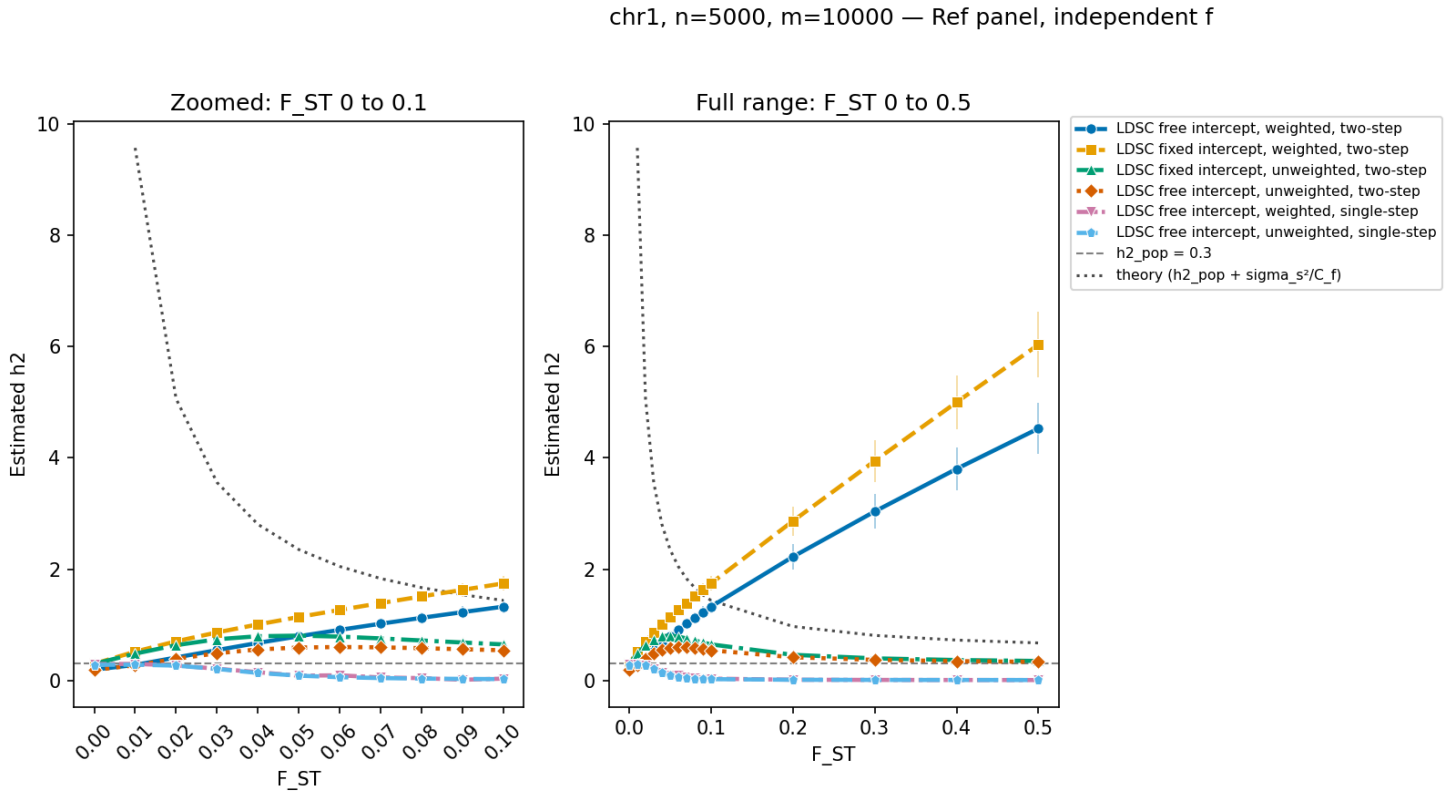
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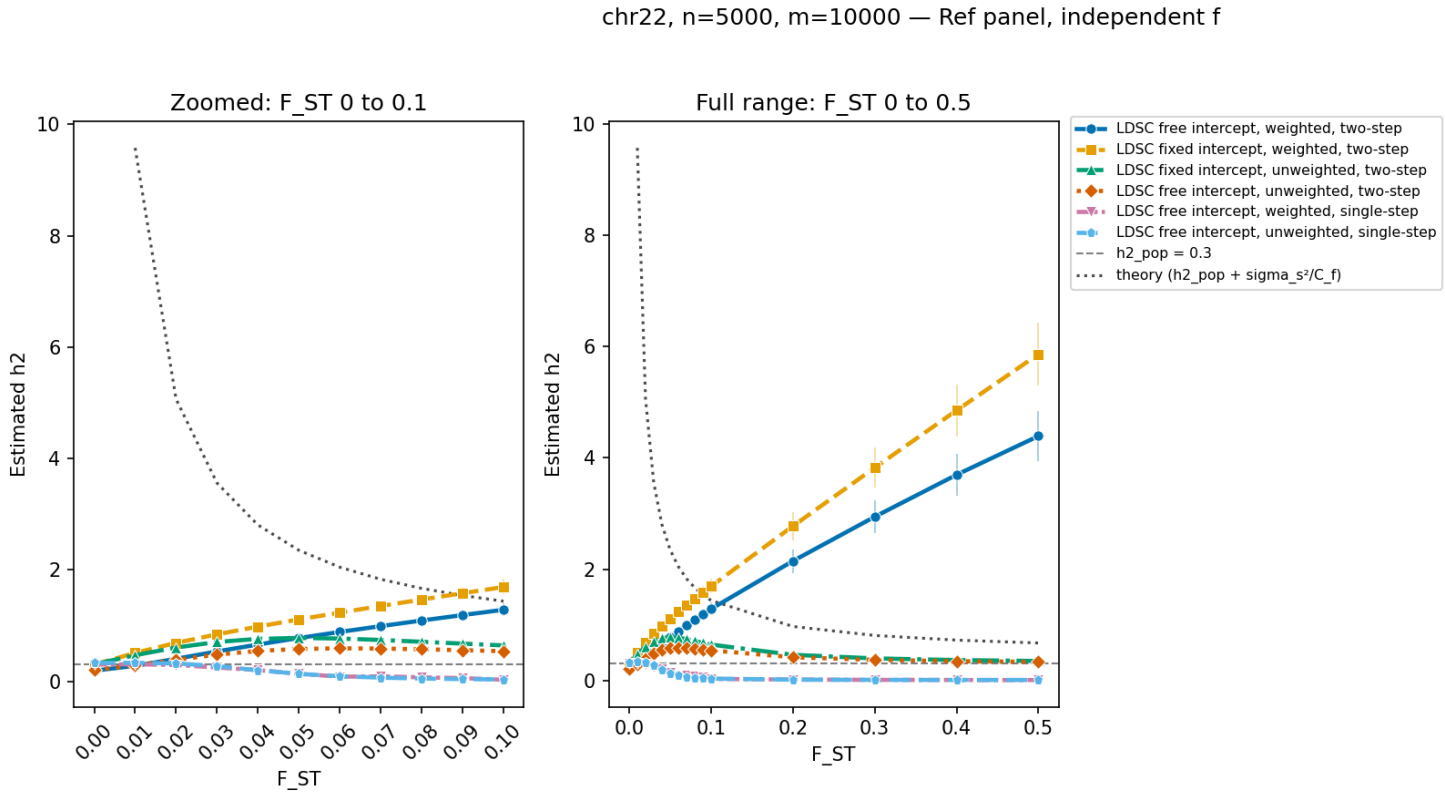
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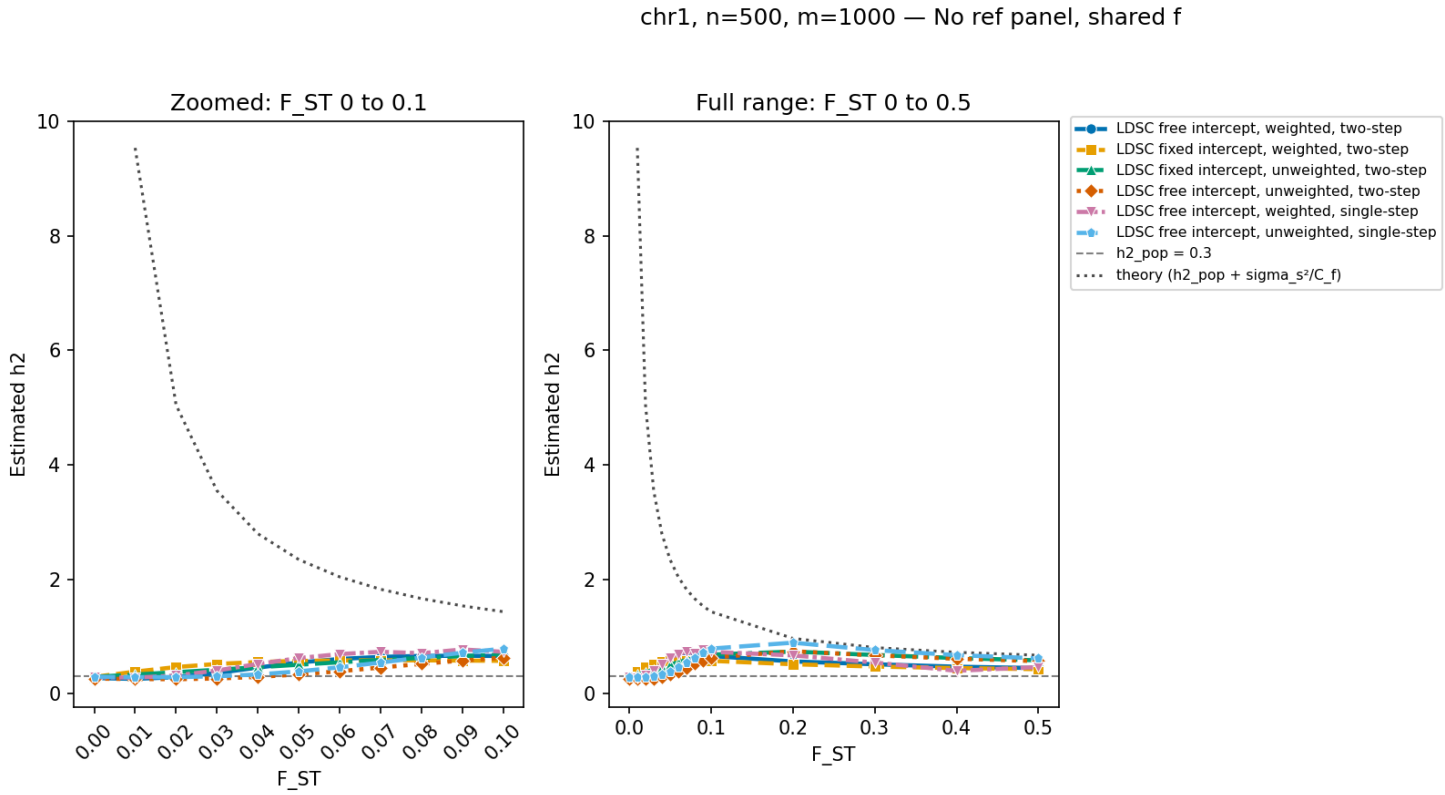
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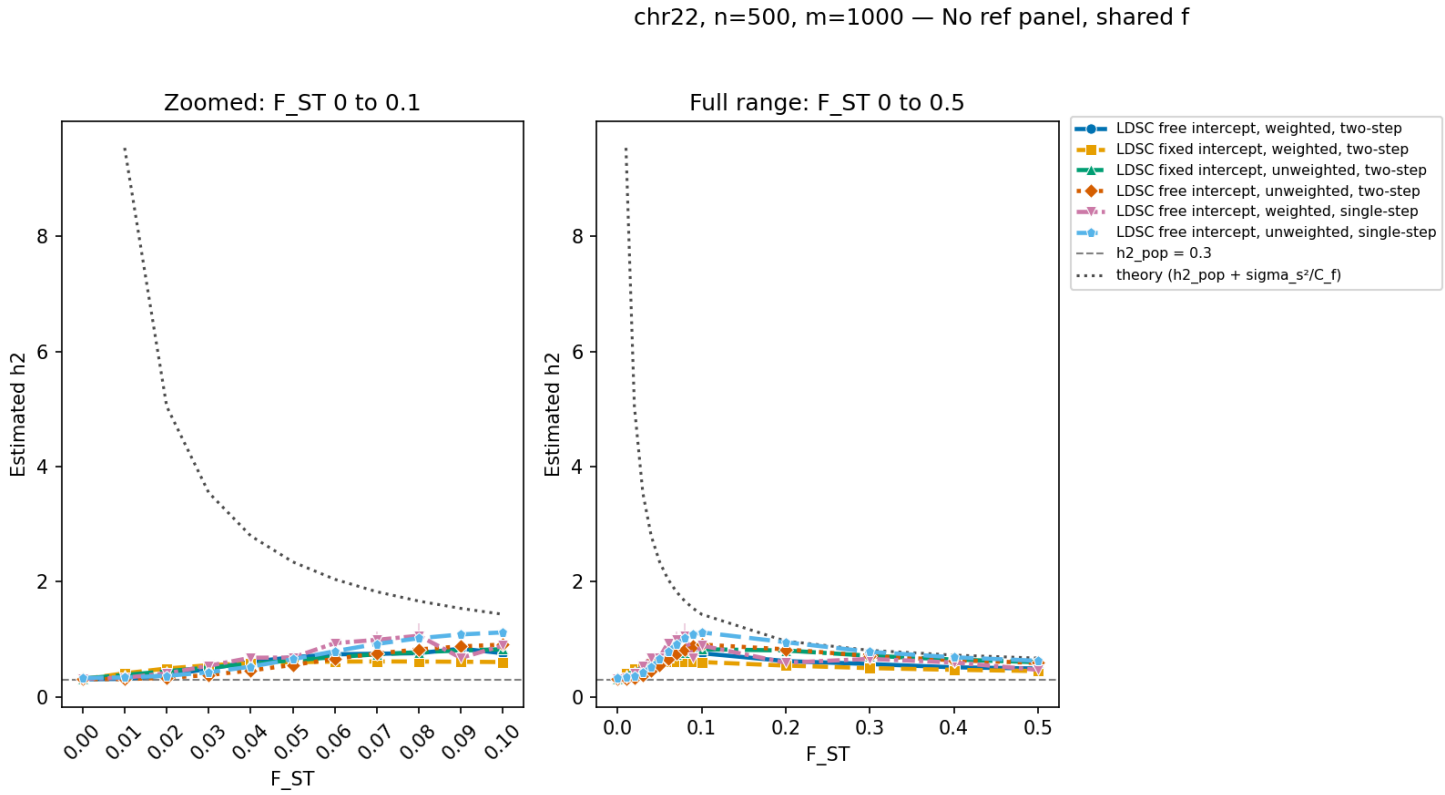
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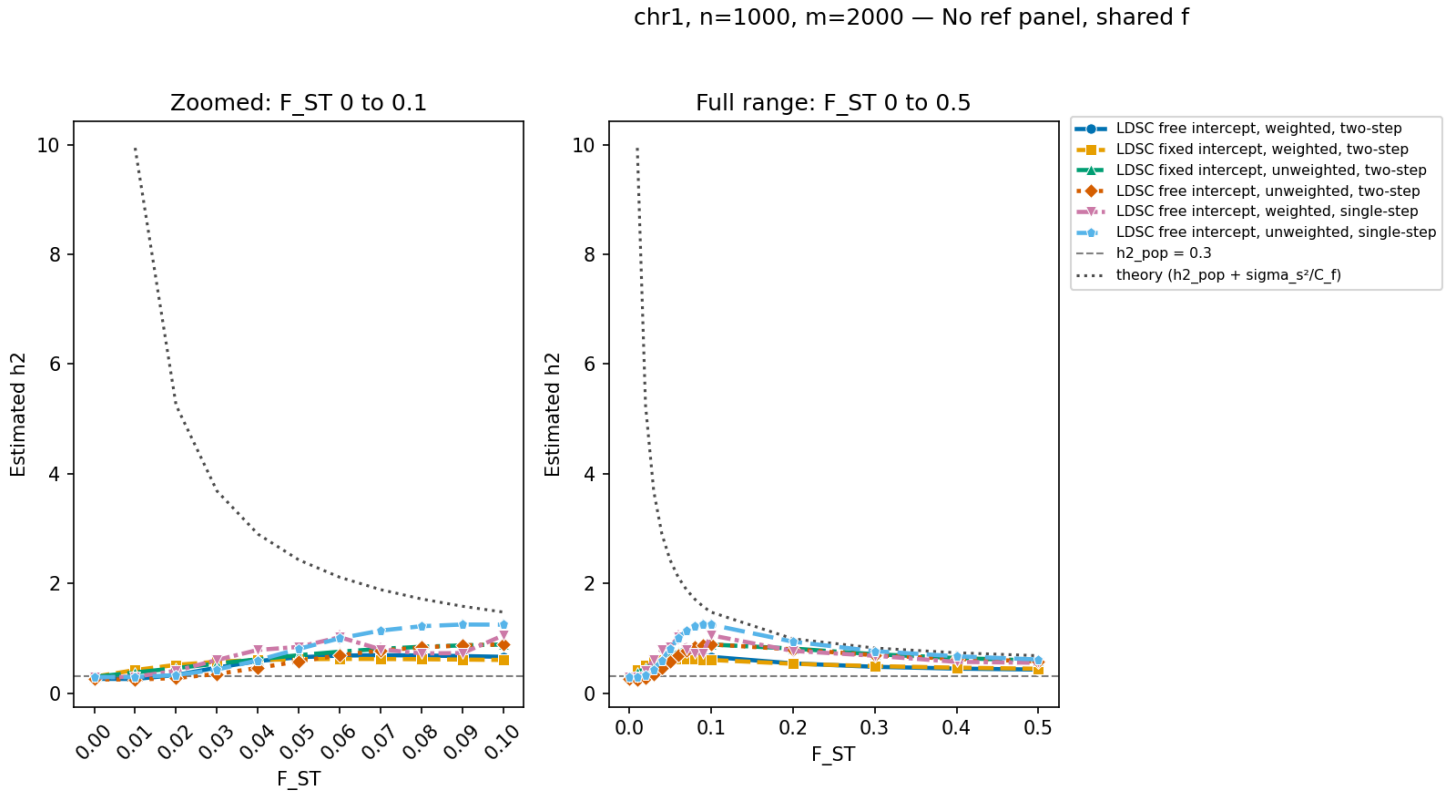
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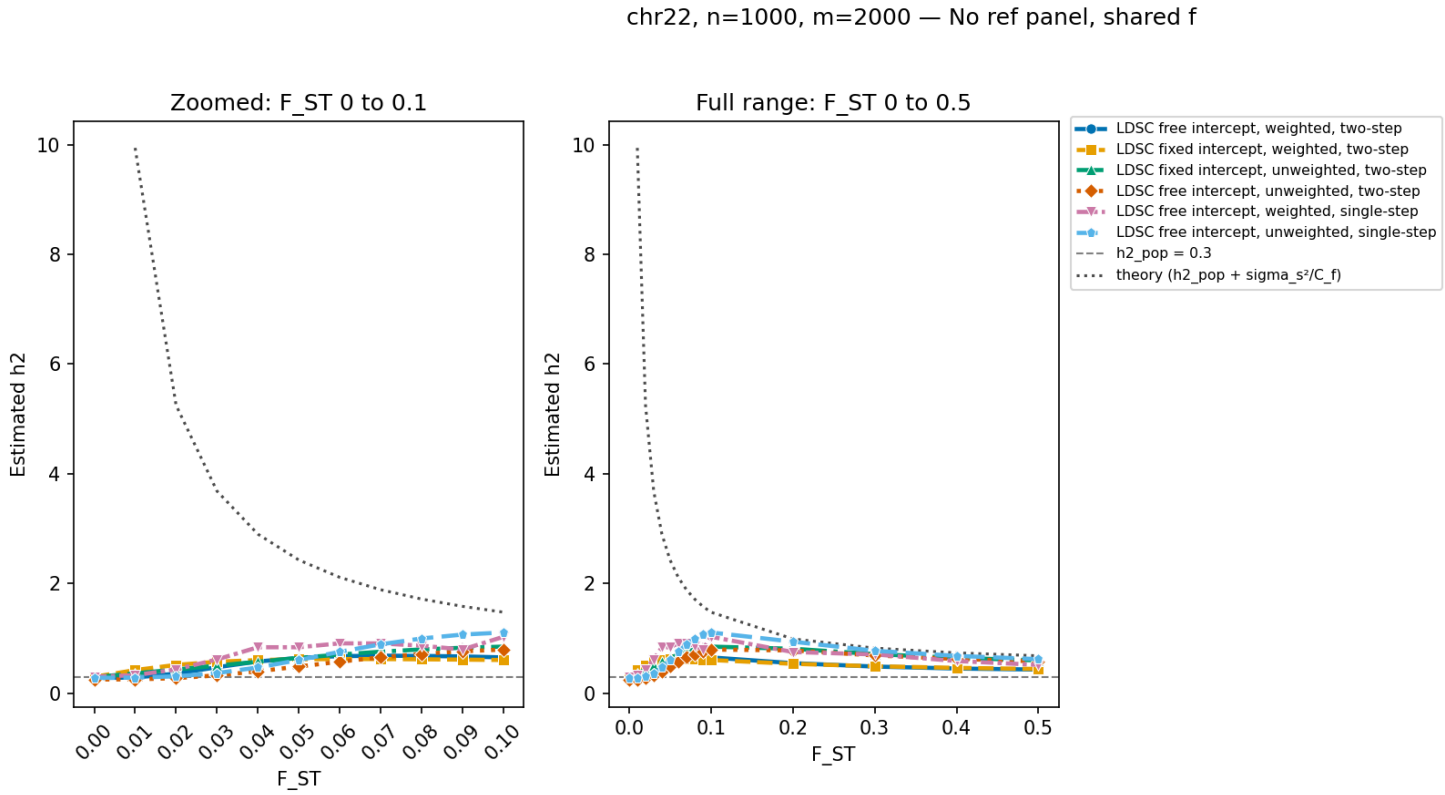
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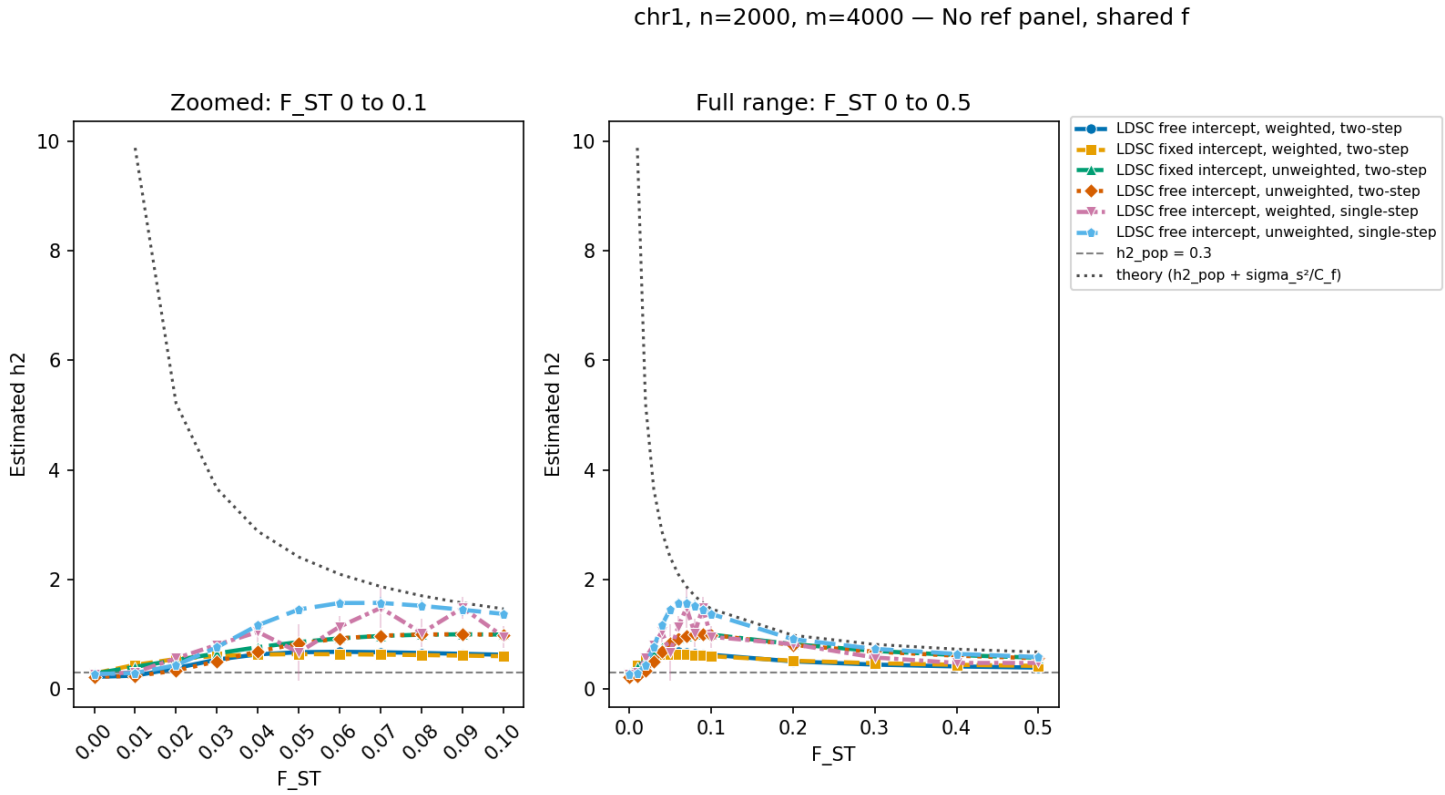
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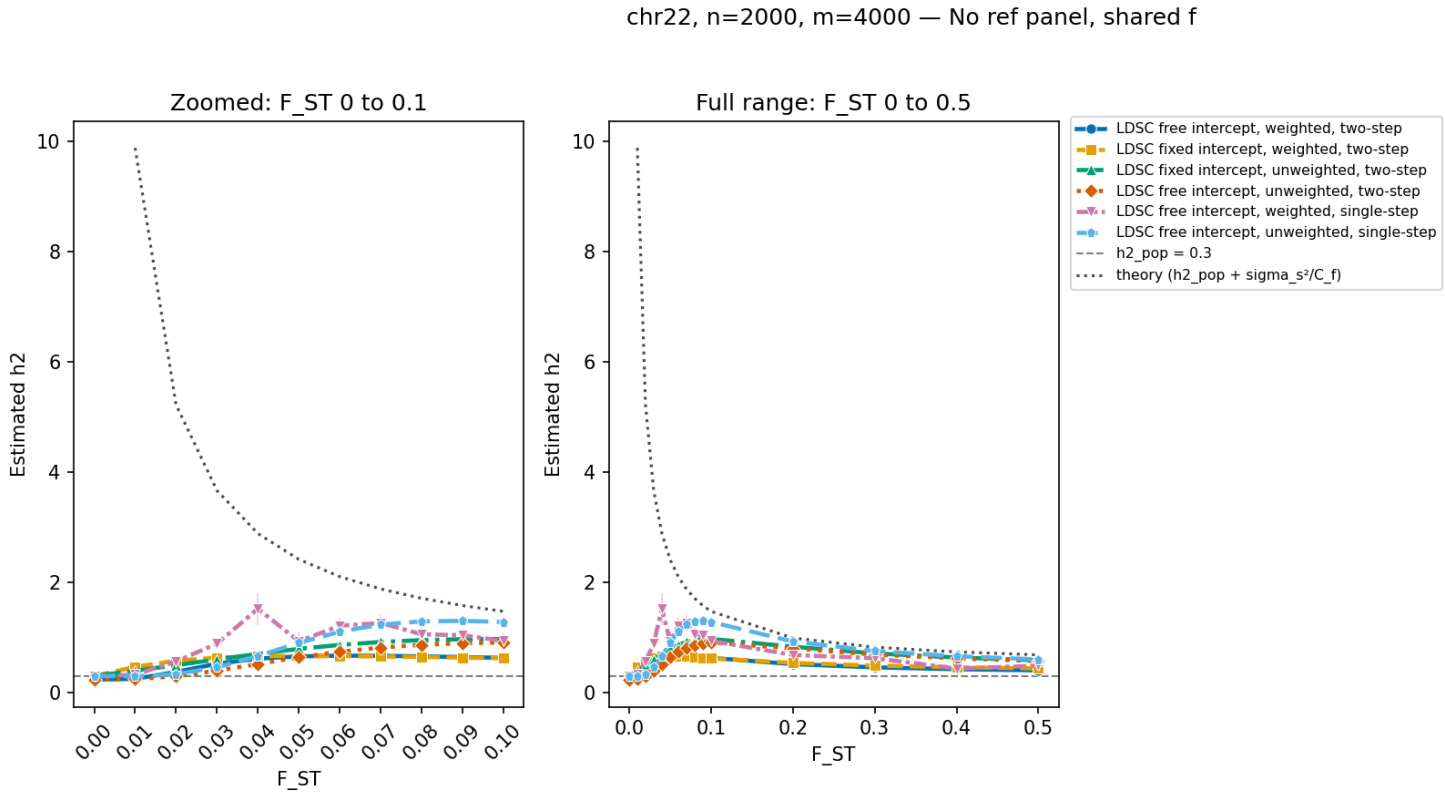
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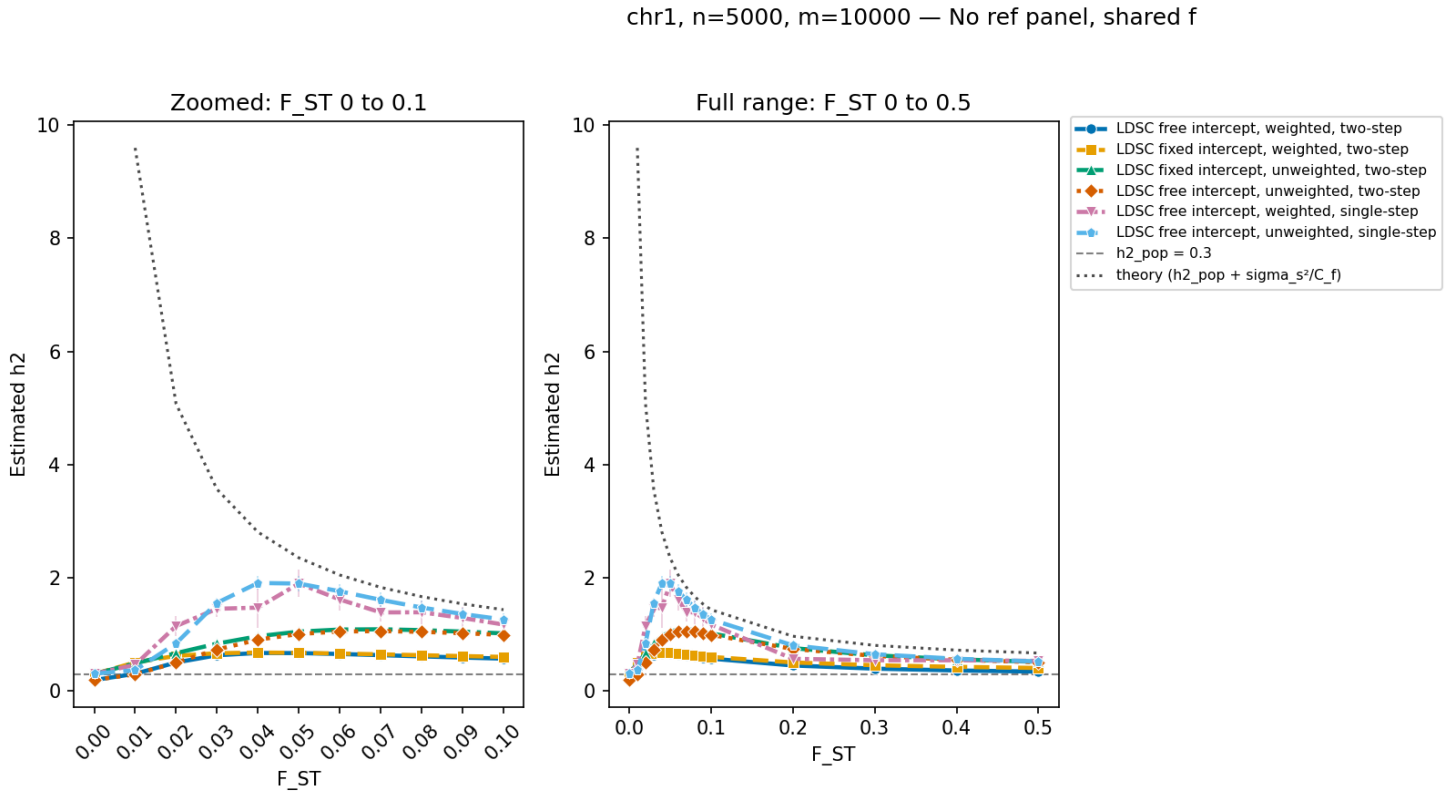
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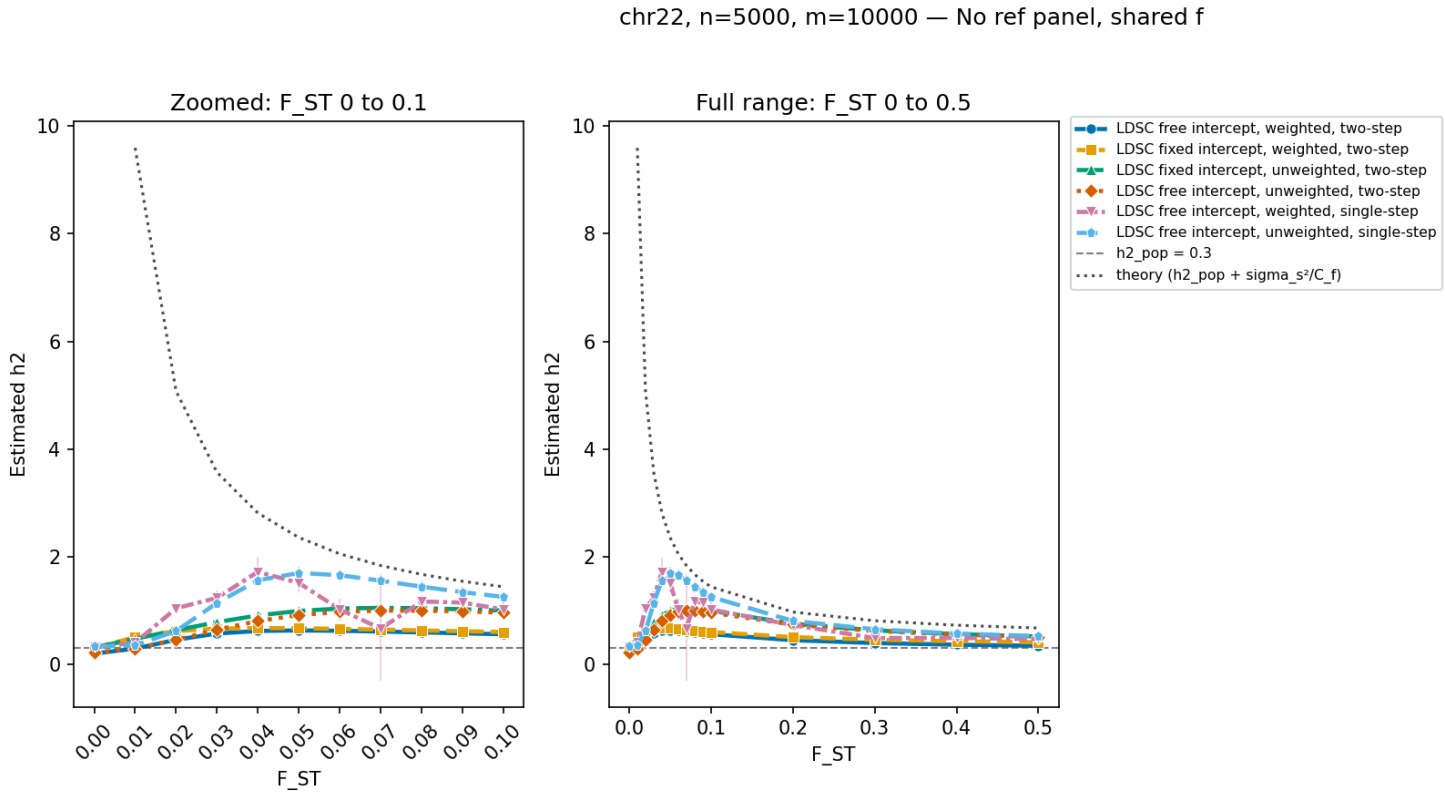
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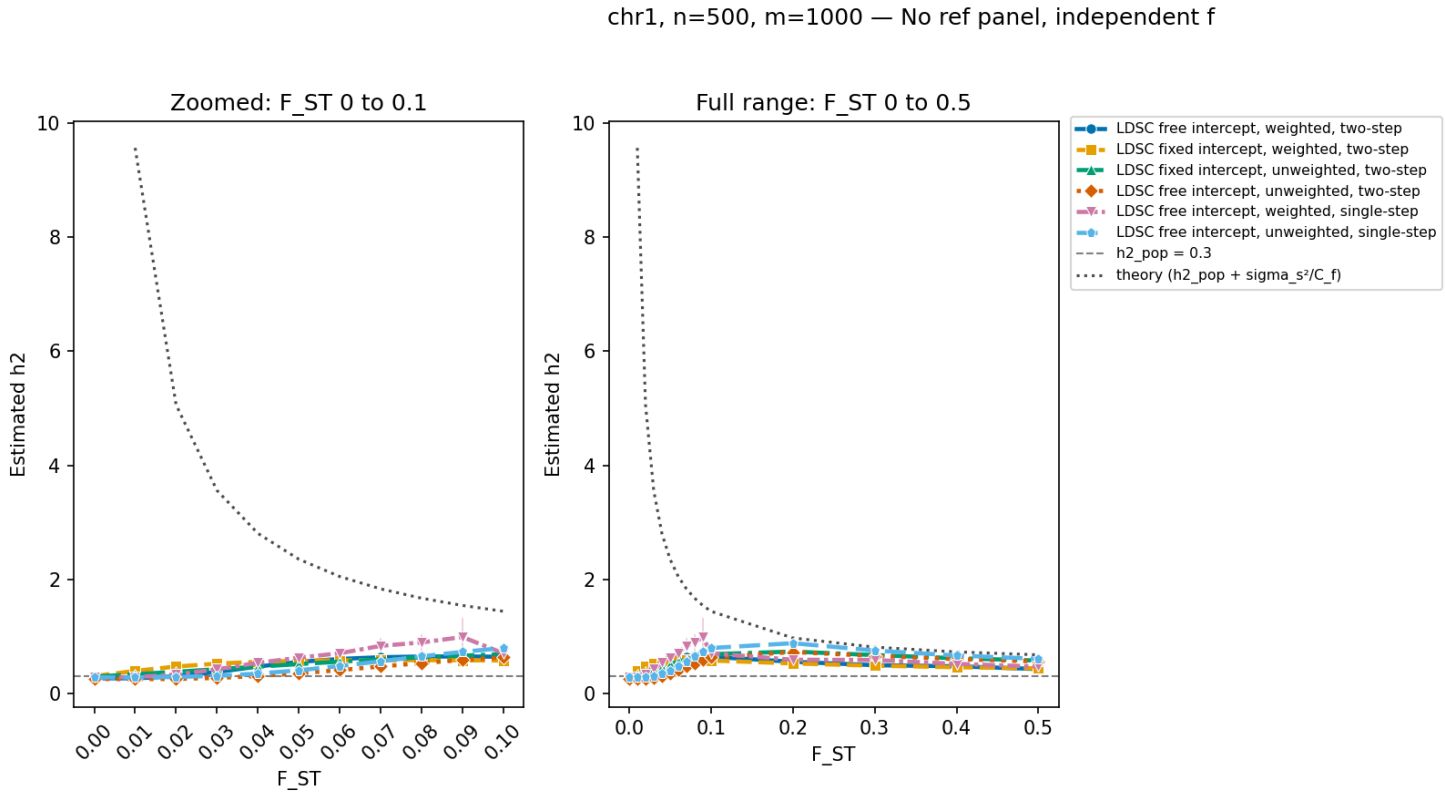
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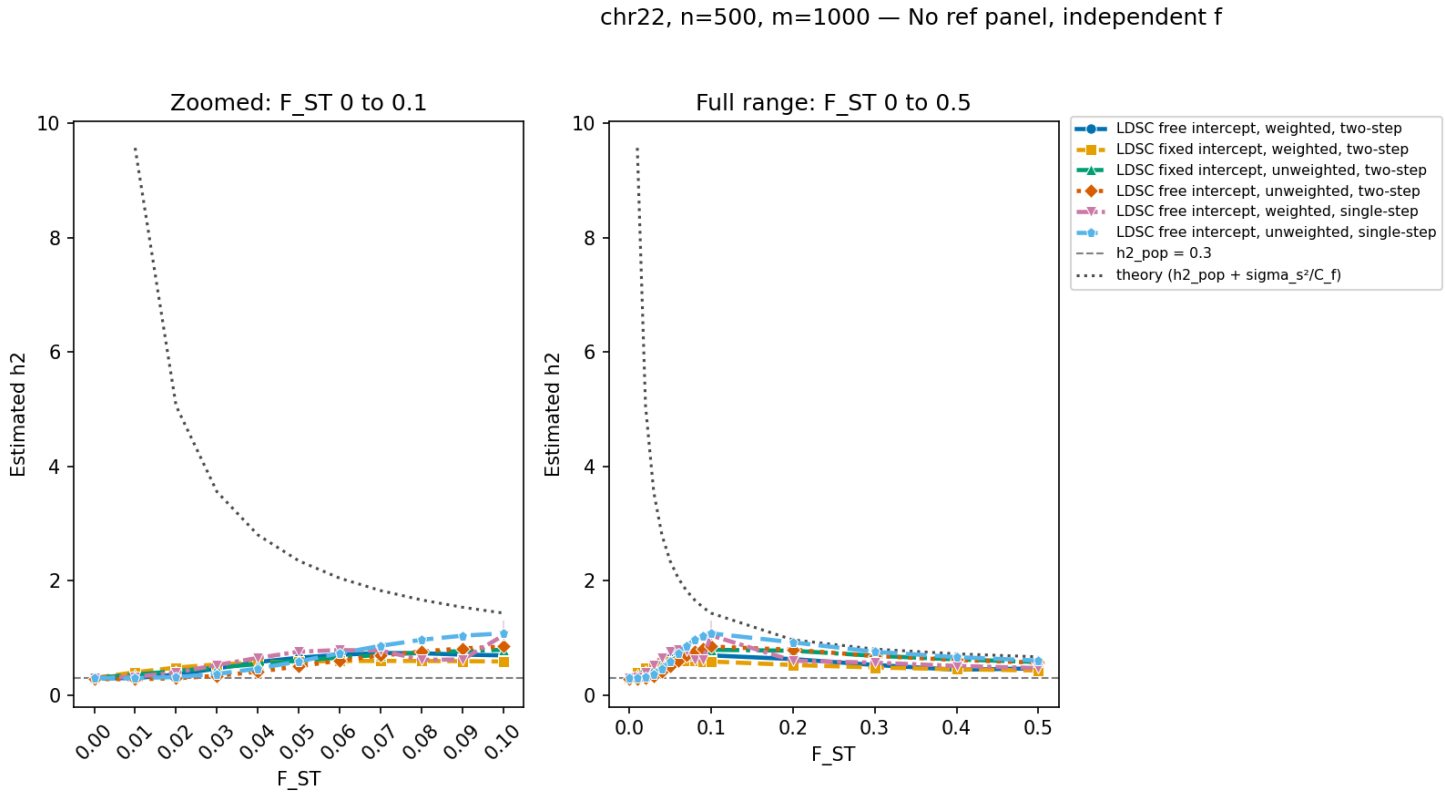
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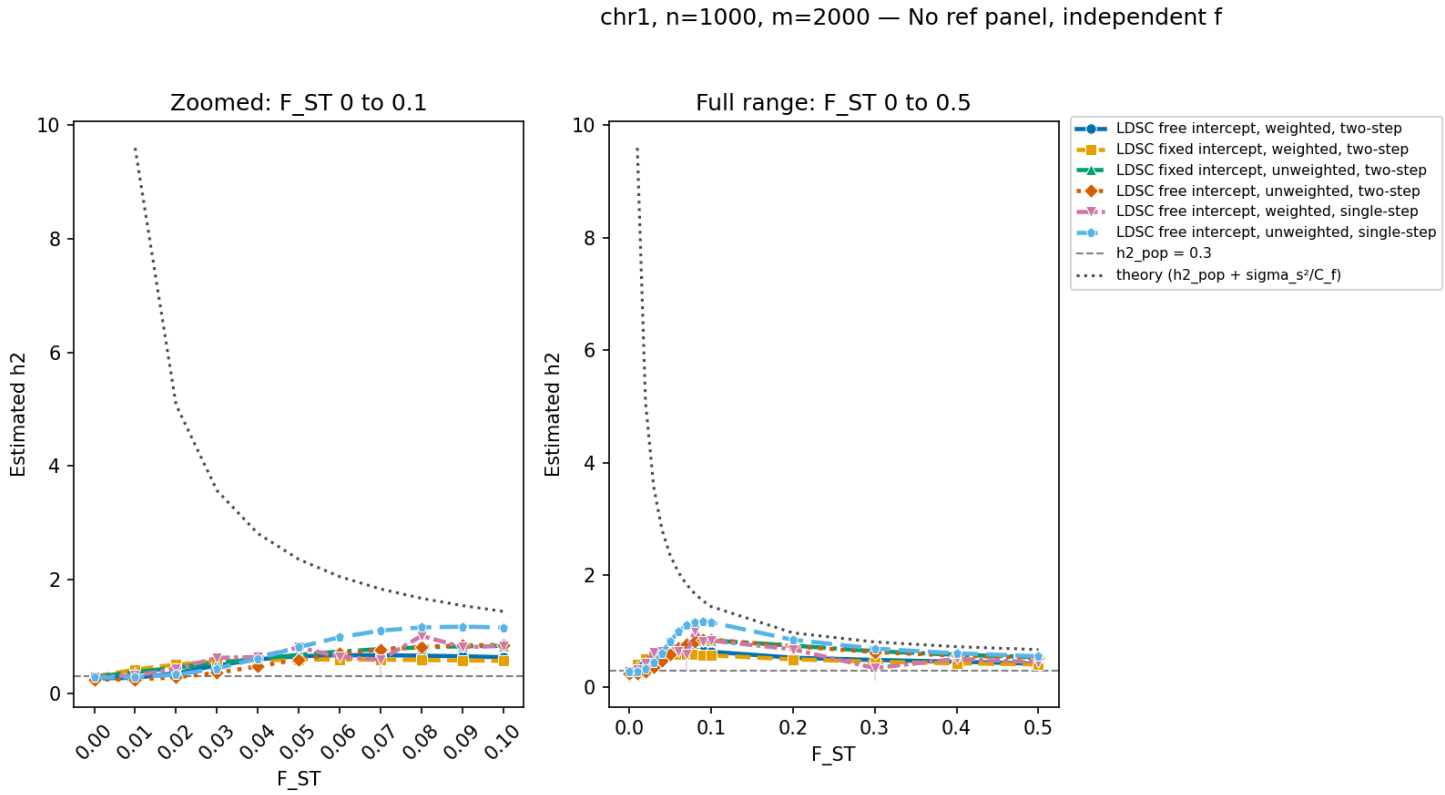
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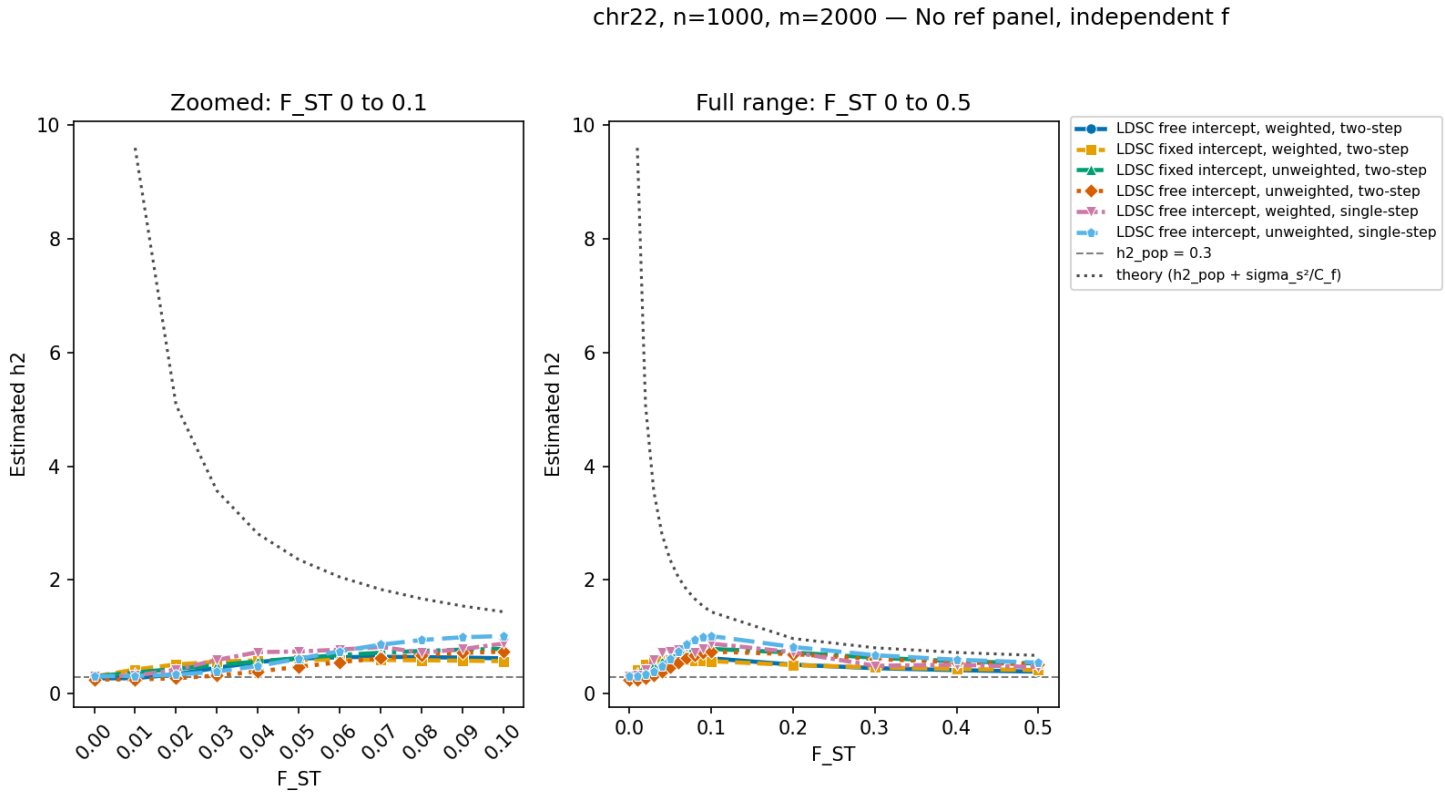
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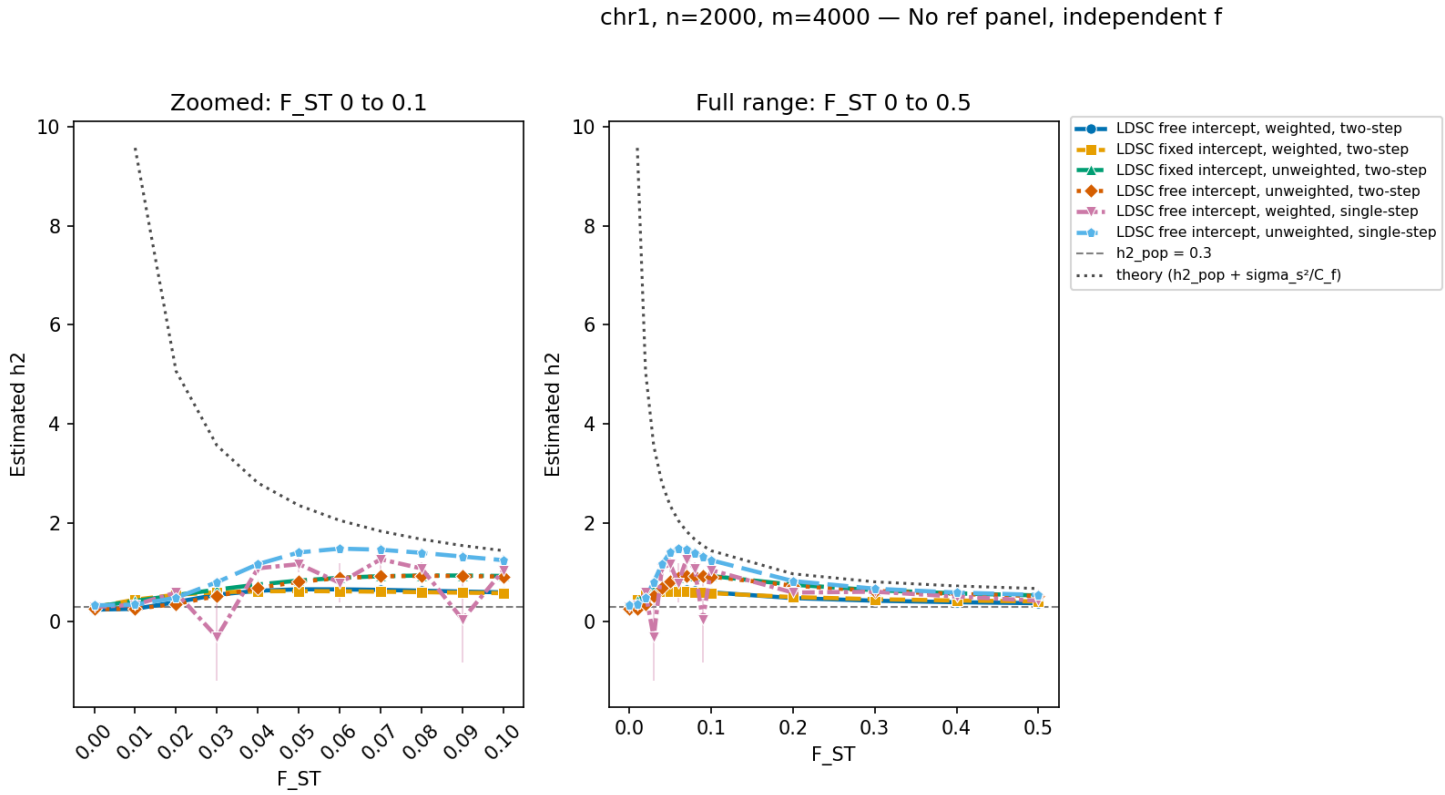
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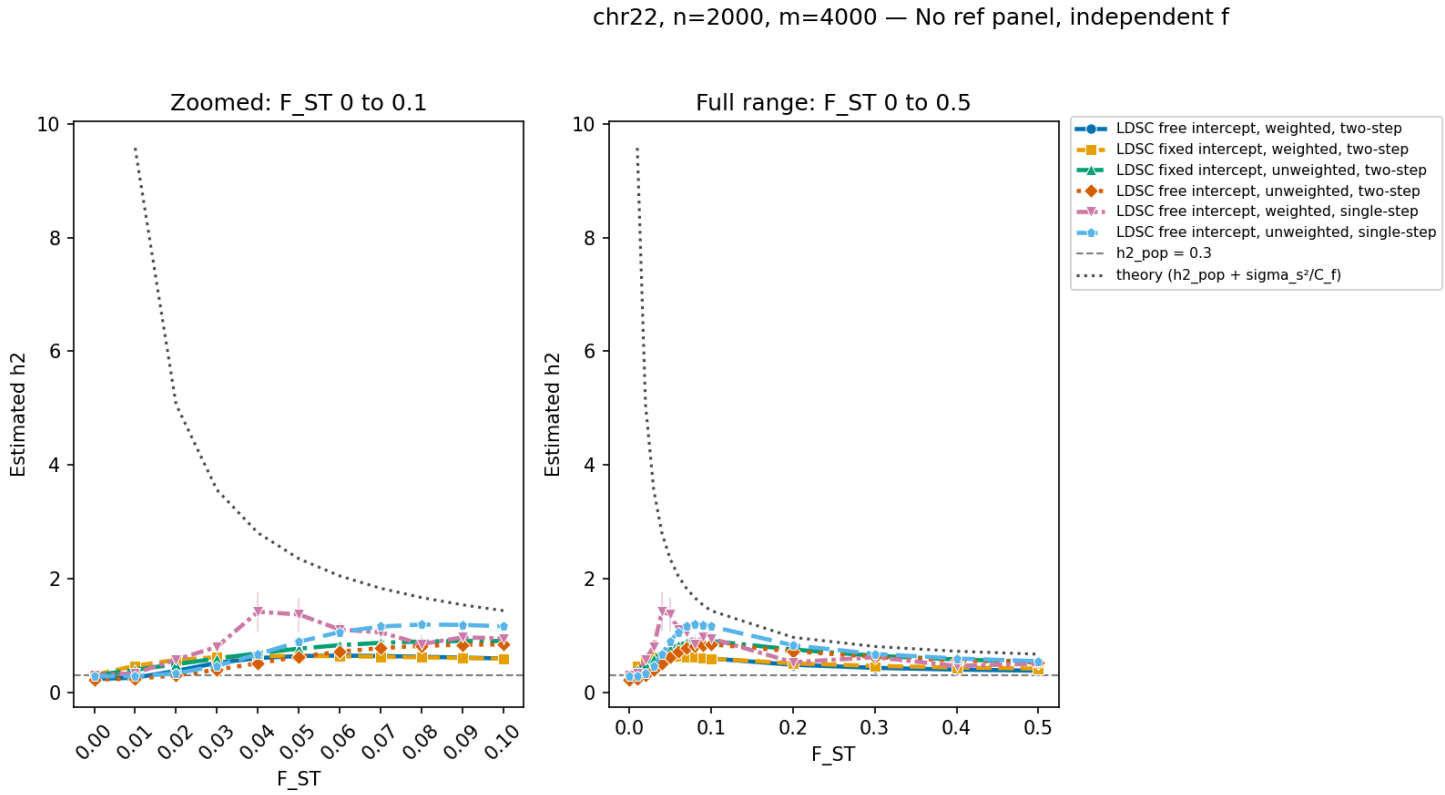
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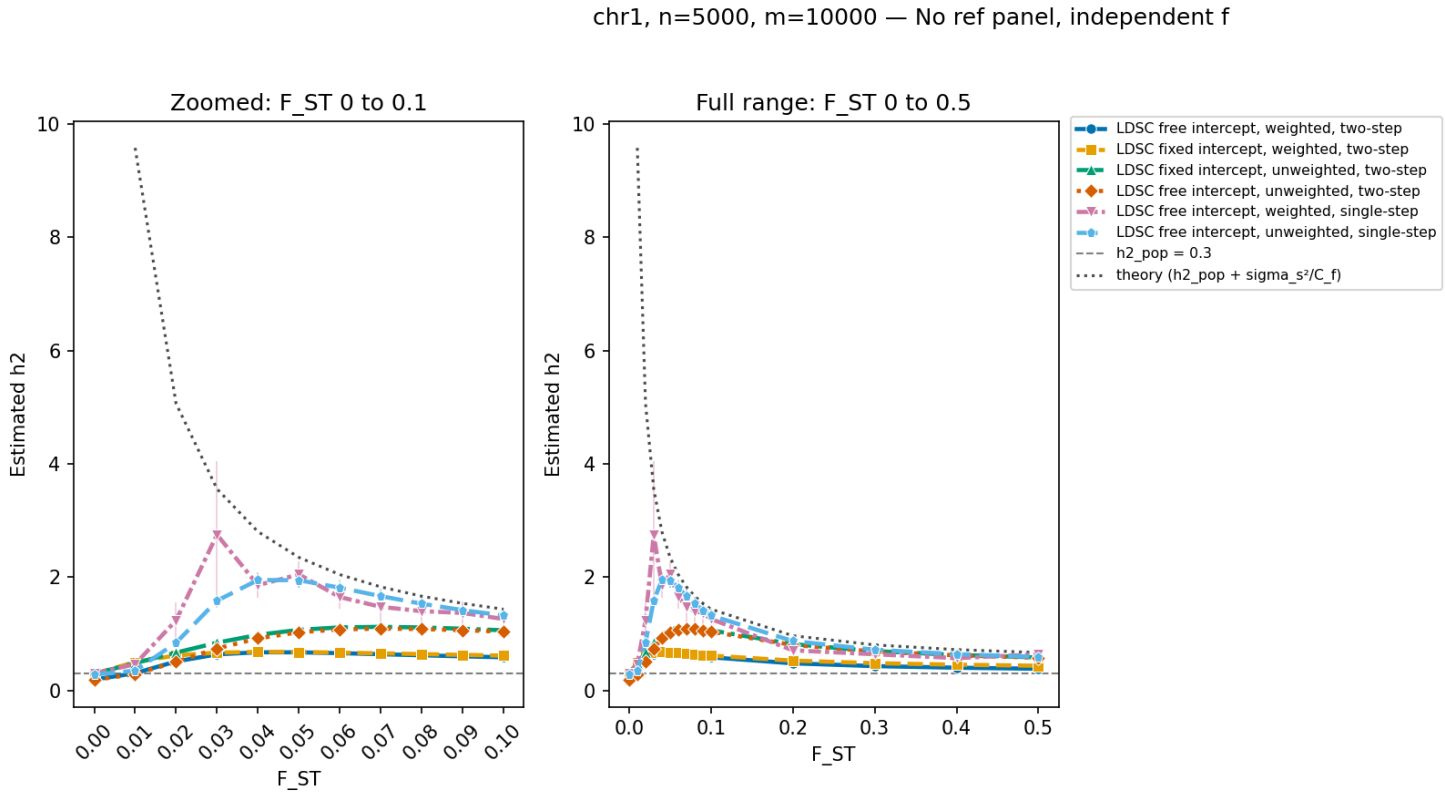
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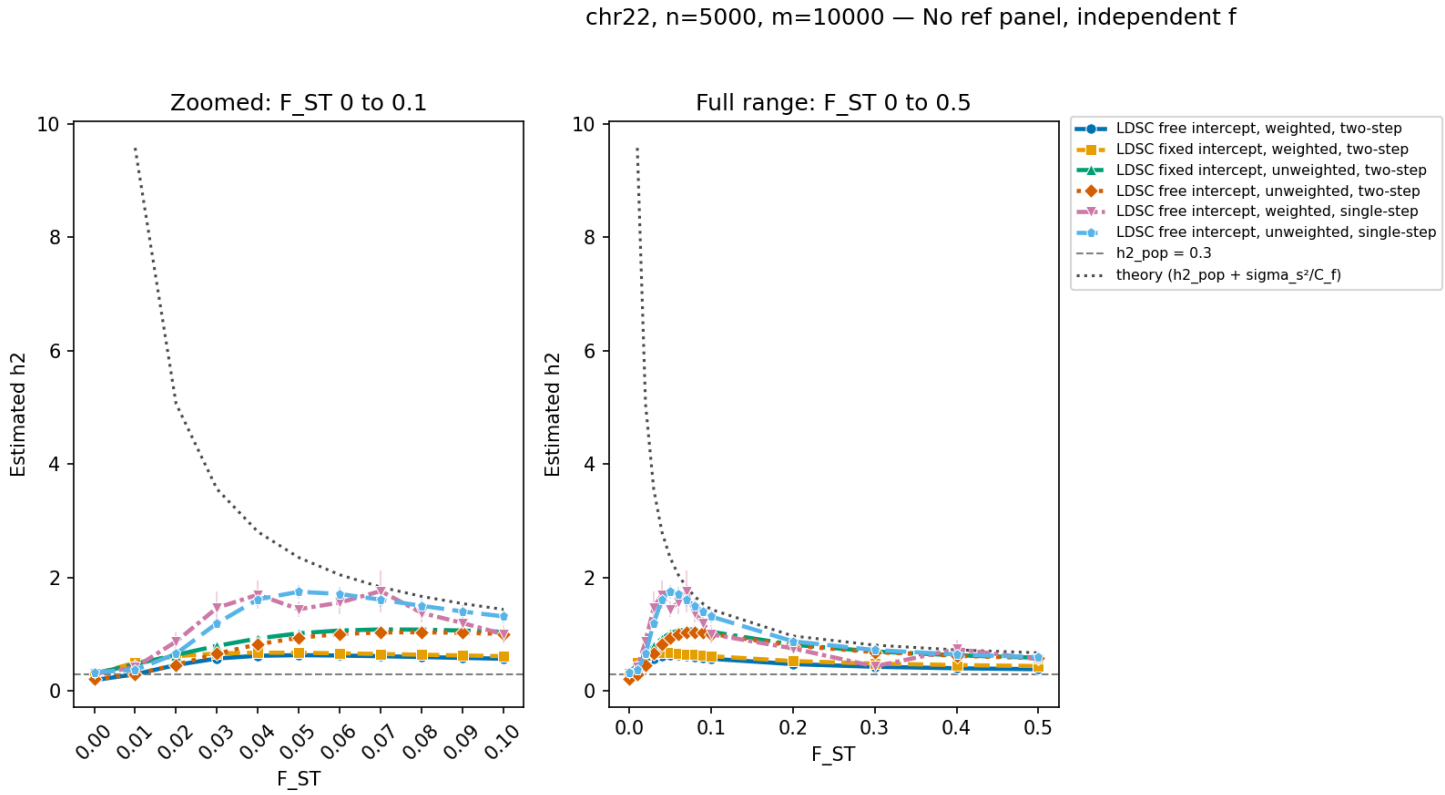
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# Discussion

**Why two-step vs. single-step matters so much:** LDSC's two-step procedure estimates the intercept first using only SNPs below a chi-square threshold, then holds that intercept fixed while re-estimating the slope (h2) using all SNPs. When the reference panel's LD scores reflect a different stratification pattern than the main sample, this two-stage process appears to compound the mismatch: the intercept absorbs some of the mismatch incorrectly, and the subsequent slope re-estimation inflates further. Skipping the two-step split (single-step) instead lets both parameters float together in one regression, which empirically drives the slope toward zero rather than away from it under this same mismatch.

**Not just an intercept or weighting story:** comparing columns pairwise - fixed vs. free intercept (both two-step, both weighted) differ only modestly (0.4-0.9 vs 0.5-1.0ish proportionally); weighted vs. unweighted (both two-step) differ more substantially but both stay bounded; it is specifically turning off the two-step procedure that produces the qualitatively different (collapsing) behavior.

**Consistent with the well-behaved scenarios:** in the three scenarios without an f-mismatched reference panel, all six variants track each other far more closely (see the figures for 'No ref panel' and 'Ref panel, shared f') - this split is not a general property of these six estimators, it is specifically triggered by the f-mismatch failure mode.

**Practical implication:** if a reference panel's stratification structure is suspected to be mismatched with the target sample, checking whether two-step and single-step LDSC estimates diverge sharply (in opposite directions) is a cheap diagnostic - in a well-specified analysis, they should not disagree this dramatically.